MATH1318 Time Series Analysis Final Project Report

SARIMA Modelling and Forecasting of Global AI Search Interest Using Google Trends Data (January 2004 – May 2026)

A Time Series Analysis Using SARIMA Modelling

TABLE OF CONTENTS

1.INTRODUCTION

Artificial Intelligence (AI) has become one of the most influential technological developments of the twenty-first century, transforming industries such as healthcare, finance, education, transportation, and software development. As AI technologies continue to evolve, public interest in the topic has increased significantly. Understanding how interest in AI changes over time can provide valuable insights into technological adoption, public awareness, and future trends.

This study analyses a monthly time series of worldwide Google Trends search interest for the term *"artificial intelligence"* from January 2004 to May 2026. Google Trends provides a normalized search index ranging from 0 to 100, where higher values indicate greater search popularity relative to the highest observed level during the study period. Because search behaviour reflects public attention and engagement, Google Trends data can serve as a useful indicator of interest in emerging technologies.

The primary research question for this study is:

"What is the most suitable time series model for forecasting worldwide interest in Artificial Intelligence, and what level of search interest can be expected over the next 10 months?"

To answer this question, several time series analysis techniques covered in the MATH1318 Time Series Analysis course are applied. The analysis begins with descriptive exploration and visualisation of the data, followed by an examination of trend, seasonality, and stationarity. Appropriate forecasting models are then developed and compared using statistical criteria and diagnostic checks. Finally, the selected model is used to generate forecasts for the next 10 months, providing insights into the future trajectory of global interest in Artificial Intelligence.

The findings of this study contribute to understanding the evolution of public interest in AI and demonstrate the application of time series modelling techniques to a contemporary real-world dataset.

2. METHODOLOGY

2.1 Data and Materials

The dataset used in this study is a public Google Trends time series measuring worldwide search interest for the term *"artificial intelligence"*. The data are recorded monthly from January 2004 to May 2026, giving a total of 269 observations.

The response variable is the Google Trends search interest index, which ranges from 0 to 100. A value of 100 represents the highest level of search interest during the selected period, while lower values represent relative search popularity compared with that peak.

The dataset was imported into R from a CSV file and converted into a monthly time series object with frequency 12. The analysis was conducted using R, with packages such as forecast, TSA, tseries, fUnitRoots, fGarch, and rugarch used for time series modelling, diagnostic testing, and forecasting.

This dataset is suitable for time series analysis because it is long enough to study changes over time and contains clear features such as trend, non-stationarity, possible seasonality, and changing volatility.

Table 1. Dataset description and structure

Item	**Value**
Source	Google Trends
Search term	artificial intelligence
Region	Worldwide
Frequency	Monthly
Period	Jan 2004 – May 2026
Observations	269
Variable	Search interest index

2.2 Analytical Pipeline

The analysis followed a structured time series modelling process.

First, the monthly Google Trends data were loaded, cleaned, and converted into a time series object. The time series was then explored using descriptive statistics and graphical analysis to identify the main characteristics of the data, including trend, seasonality, outliers, and changes in variability.

Second, the series was assessed for stationarity. Visual inspection, ACF and PACF plots, and formal stationarity tests were used to determine whether differencing or transformation was required. Since the original series showed strong non-stationary behaviour, transformations and differencing were considered before fitting ARIMA-based models.

Third, several model classes were fitted and compared. Trend regression models were used to capture long-term growth patterns. ARIMA models were then fitted to model autocorrelation in the non-seasonal structure of the series. SARIMA models were also considered because the data are monthly and may contain seasonal dependence. In addition, GARCH-type models were examined to investigate whether the series showed changing volatility over time.

Fourth, fitted models were evaluated using diagnostic checking. Residual plots, ACF plots of residuals, normality checks, and Ljung-Box tests were used to assess whether the residuals behaved like white noise.

Finally, competing models were compared using AIC and BIC values, together with forecast performance and residual diagnostics. The selected model was then used to forecast worldwide AI search interest for the next 10 months, from June 2026 to March 2027.

3. RESULTS

3.1 Descriptive Analysis

Table 2 presents summary statistics for the monthly Google Trends search interest series for the term *"artificial intelligence"* from January 2004 to May 2026. The series has a mean value of 13.11 and a median of 9, indicating that most observations remained relatively low throughout much of the study period. However, the maximum value of 100 and the large difference between the median and maximum suggest the presence of substantial growth in recent years.

Table 2. Summary statistics for AI search interest

Statistic	**Value**
Minimum	4.00
1st Quartile	5.00
Median	9.00
Mean	13.11
3rd Quartile	14.00
Maximum	100.00

The time series plot shown in Figure 1 reveals a clear long-term increase in global interest in Artificial Intelligence. Search interest declined gradually between 2004 and approximately 2010 before remaining relatively stable at low levels for several years. From around 2016 onwards, interest began increasing steadily, followed by a sharp acceleration after late 2022. The timing of this acceleration coincides with the public release of ChatGPT in November 2022, marked by the vertical reference line in Figure 1. Search activity reached unprecedented levels during 2025 and 2026, indicating rapidly increasing public attention towards AI-related technologies.

Figure 1. Monthly Global Search Interest in 'Artificial Intelligence' (Google Trends, Jan 2004 – May 2026)

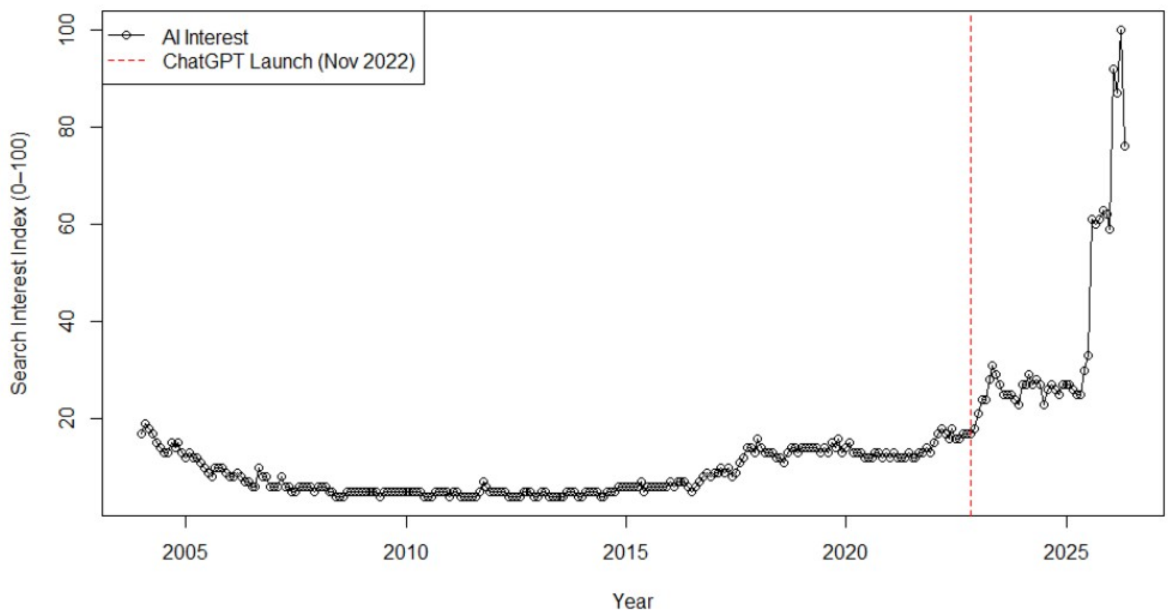
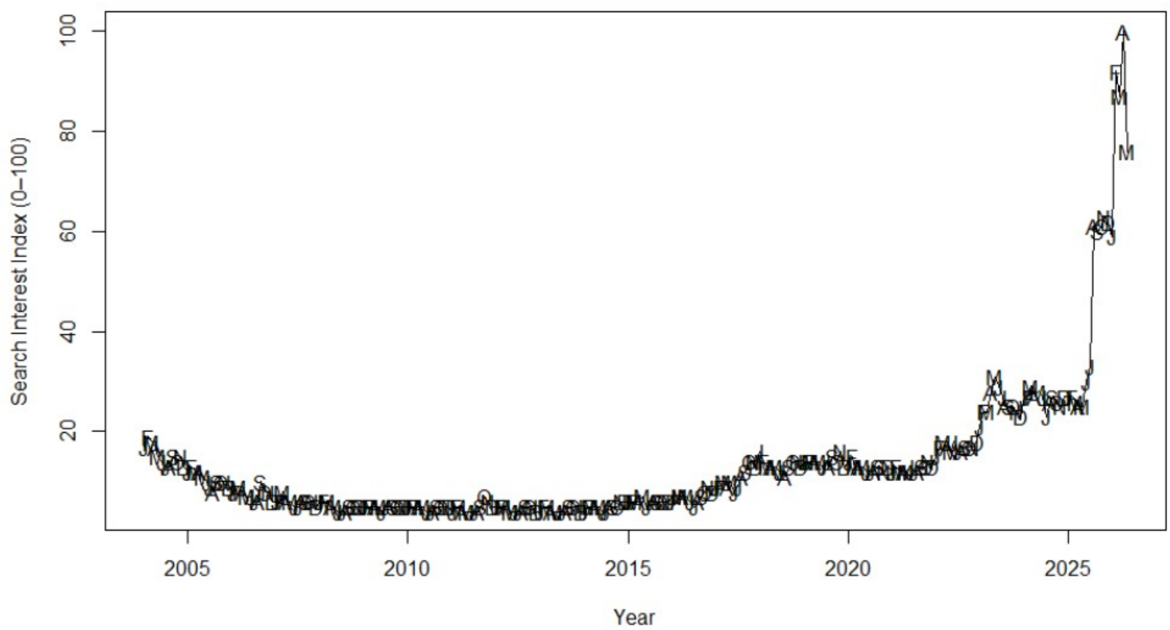


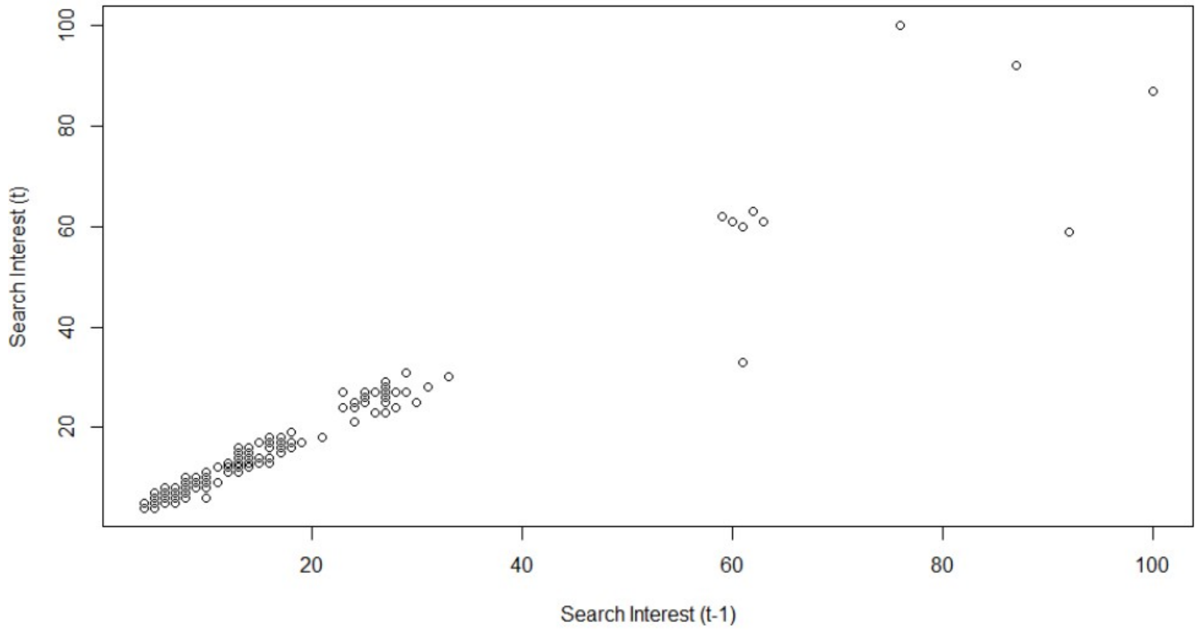
Figure 2 presents the monthly observations labelled by month. Although minor fluctuations are visible throughout the year, there is no strong or consistent seasonal pattern apparent from visual inspection. Instead, the dominant feature of the series is the substantial upward trend observed in recent years.

Figure 2. Monthly Search Interest with Seasonal Labels



The lag-1 scatter plot in Figure 3 shows a very strong positive relationship between consecutive observations. The estimated lag-1 correlation is 0.970, indicating that search interest in one month is highly associated with search interest in the previous month. This level of persistence is typical of highly dependent time series and suggests substantial temporal autocorrelation.

Figure 3. Lag-1 Scatter Plot



The autocorrelation function (ACF) shown in Figure 4 exhibits a slow decay across many lags, with significant positive autocorrelations extending well beyond one year. Such behaviour is characteristic of a non-stationary series containing a strong trend component. Similarly, the partial autocorrelation function (PACF) in Figure 5 displays a large spike at lag 1 followed by smaller but still notable correlations at subsequent lags.

Figure 4. ACF of AI Search Interest

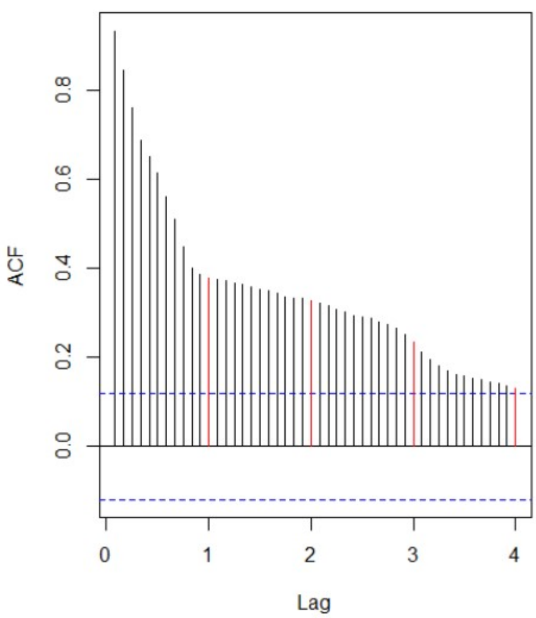
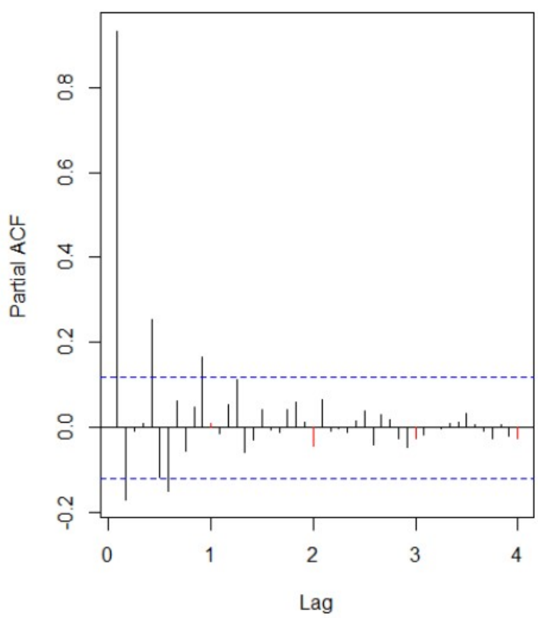


Figure 5. PACF of AI Search Interest



Overall, the descriptive analysis indicates that the AI search interest series is characterised by strong persistence, substantial long-term growth, and possible weak seasonal effects. These findings suggest that simple descriptive methods are insufficient for forecasting and motivate the use of more advanced time series models in the following sections.

3.2 Trend Models

Several regression-based trend models were fitted to capture the long-term behaviour of the AI search interest series. These included a linear trend model, a quadratic trend model, a cubic trend model, a seasonal means model, a seasonal plus quadratic trend model, and a harmonic seasonal model.

Figure 6 presents the fitted linear trend model. The model identified a statistically significant positive trend over time; however, visual inspection shows that the fitted line fails to capture the rapid acceleration in search interest observed during the later years of the study period. The model achieved an adjusted R^2 value of 0.344, indicating that only a moderate proportion of the variation in the series was explained by a simple linear trend.

Figure 6. Linear Trend Model

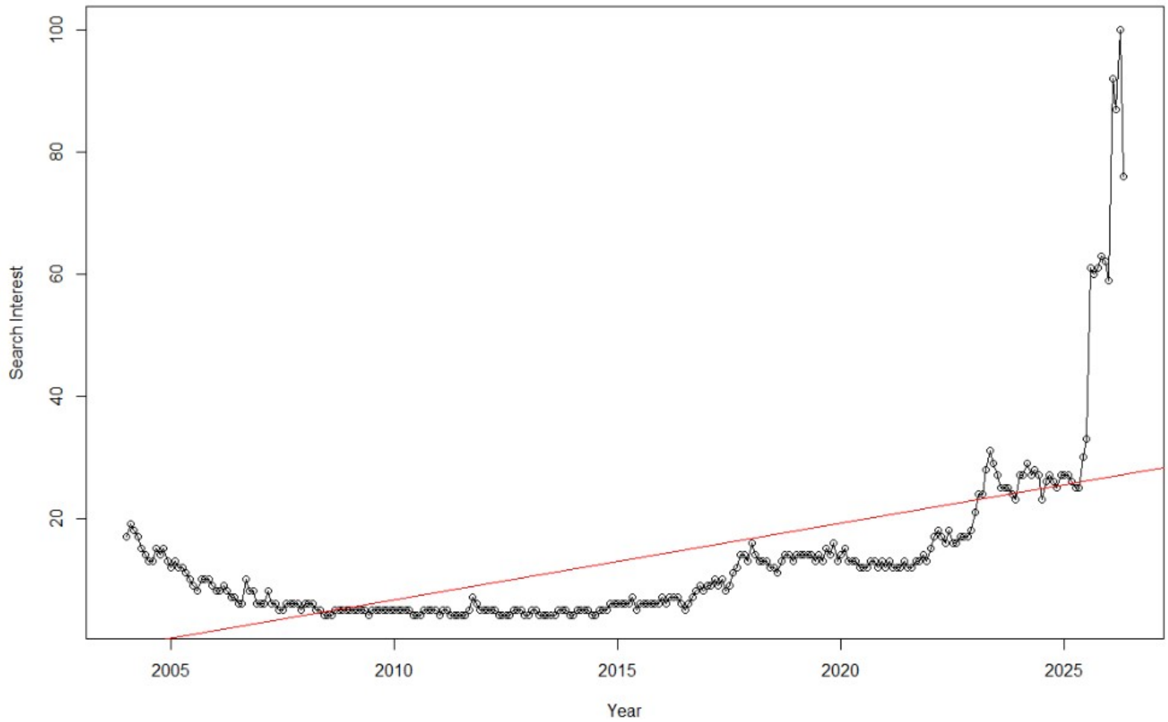
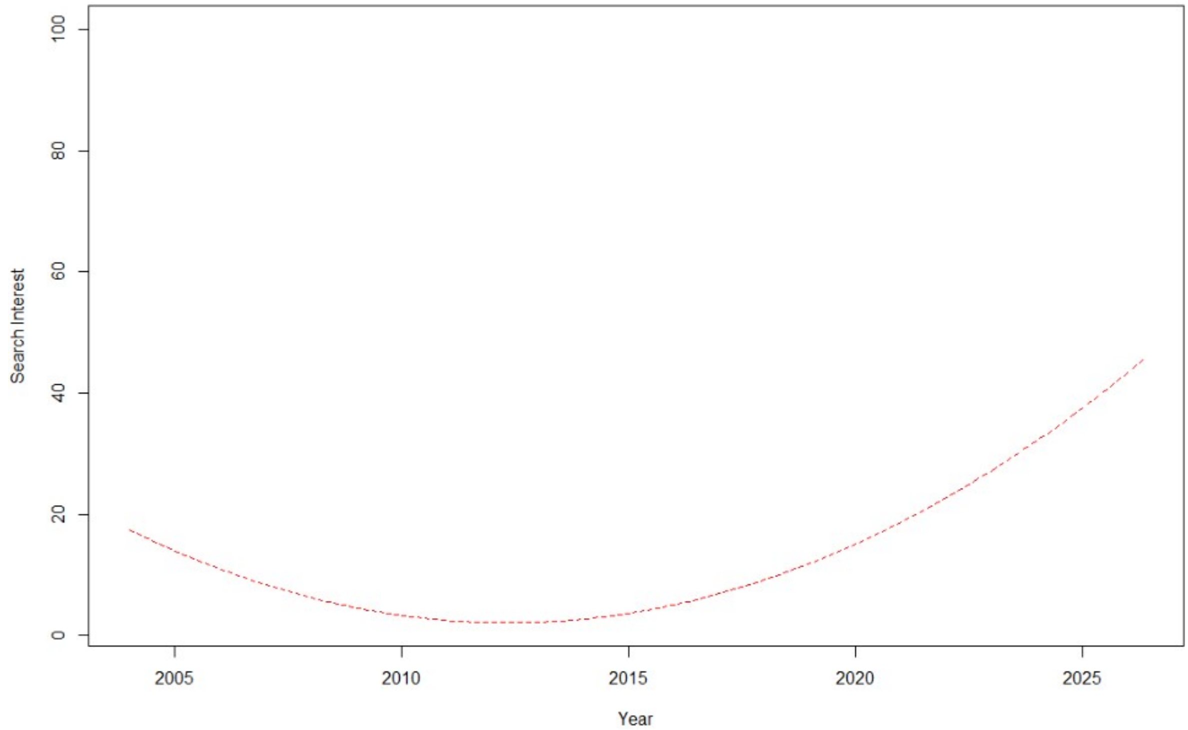


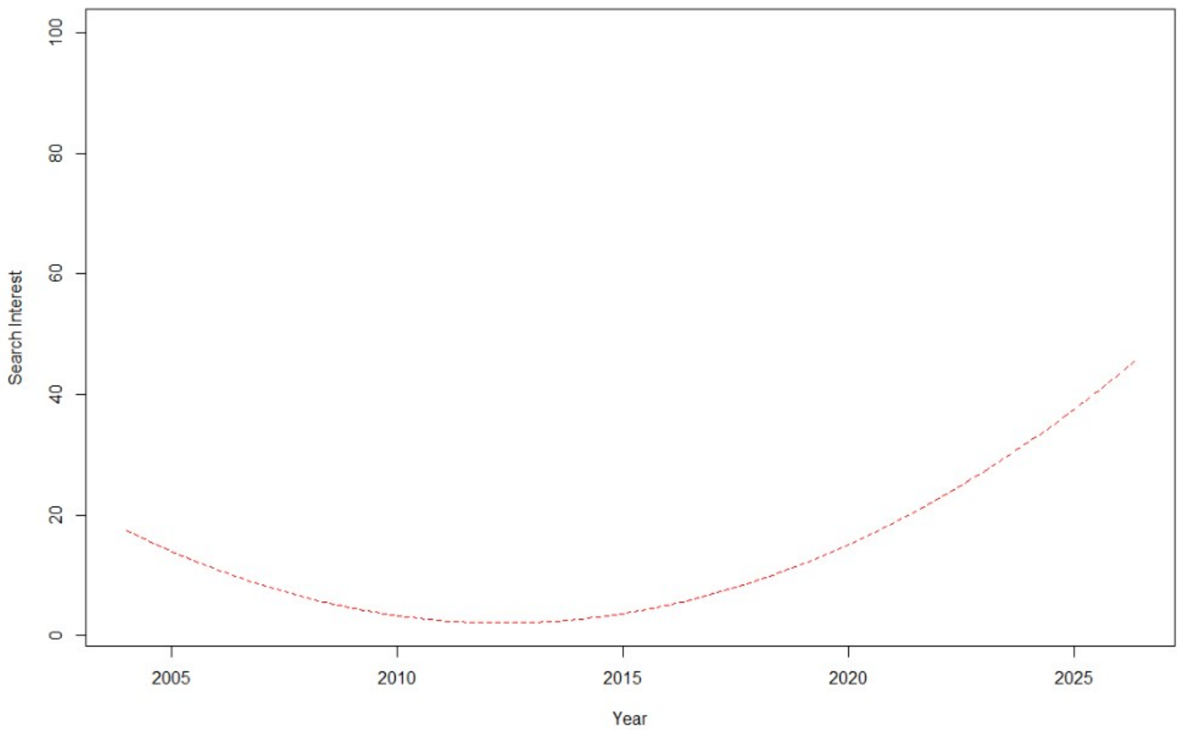
Figure 7 shows the quadratic trend model. By allowing the growth rate to change over time, the quadratic specification provided a substantially better fit to the data than the linear model. The adjusted R^2 increased to 0.706, demonstrating that the nonlinear trend captured much of the long-term behaviour of the series. The fitted curve closely reflects the gradual decline in the early years followed by the strong increase observed after 2015.

Figure 7. Quadratic Trend Model



A cubic trend model was also estimated (Figure 8). However, the cubic term was found to be redundant and could not be estimated separately from the quadratic component. As a result, the cubic model produced identical goodness-of-fit statistics to the quadratic model and offered no additional explanatory power.

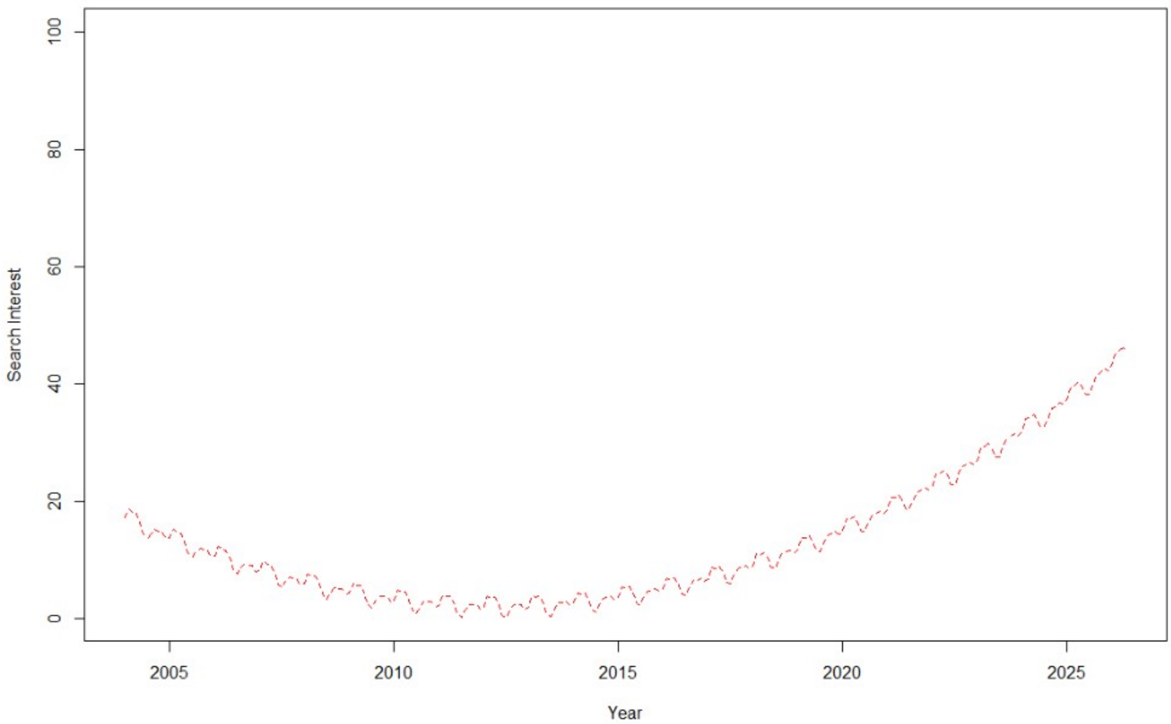
Figure 8. Cubic Trend Model



The seasonal means model considered only monthly seasonal effects. Although several monthly coefficients were statistically significant, the model achieved an adjusted R^2 of only 0.457, indicating that seasonality alone was unable to explain the dominant patterns in the data.

The best-performing regression model was the Seasonal + Quadratic Trend Model shown in Figure 9. This model combines both seasonal effects and a nonlinear time trend, resulting in an adjusted R^2 of 0.842. The fitted values closely follow the observed trajectory of the series throughout the study period, including the rapid increase in AI search interest during recent years.

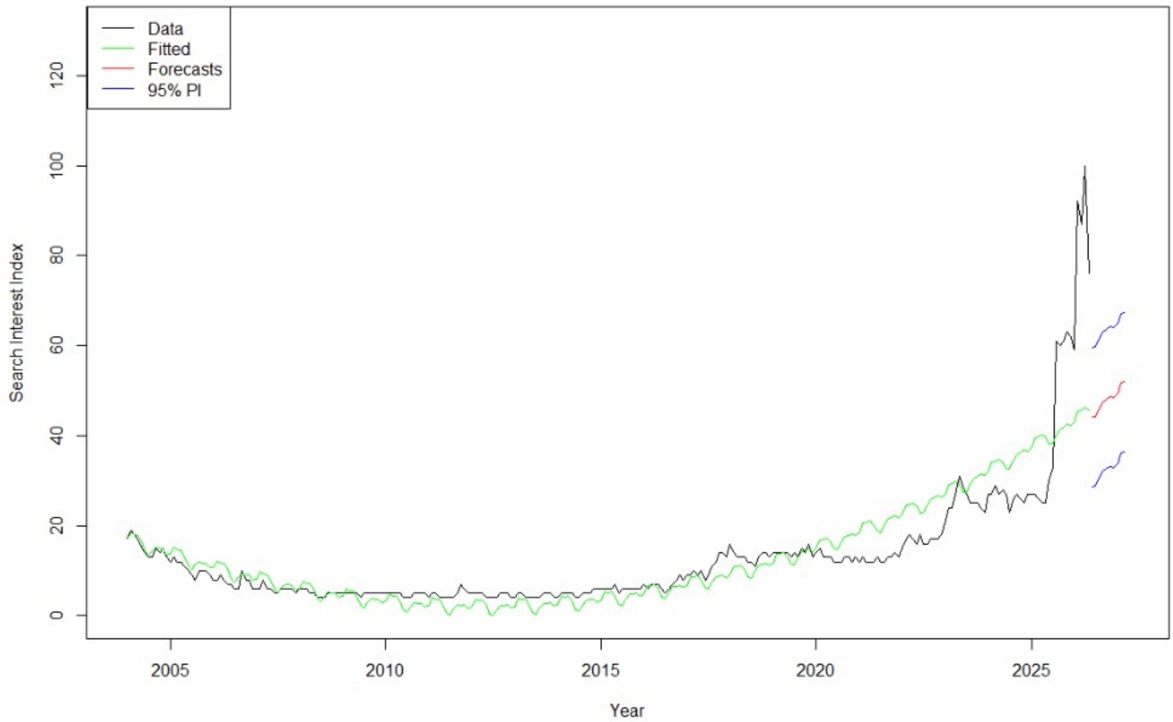
Figure 9. Seasonal + Quadratic Trend Model



Model selection was performed using the Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC). The quadratic and cubic models achieved the lowest information criterion values among the pure trend models, while the Seasonal + Quadratic model provided the best overall balance between goodness-of-fit and explanatory capability. Consequently, the Seasonal + Quadratic Trend Model was selected as the preferred regression-based forecasting model.

Using this model, forecasts were generated for the next ten months, covering June 2026 to March 2027. As shown in Figure 10, the forecasts suggest that global interest in Artificial Intelligence is expected to remain elevated and continue increasing over the forecast horizon. The widening prediction intervals indicate increasing uncertainty as the forecast extends further into the future.

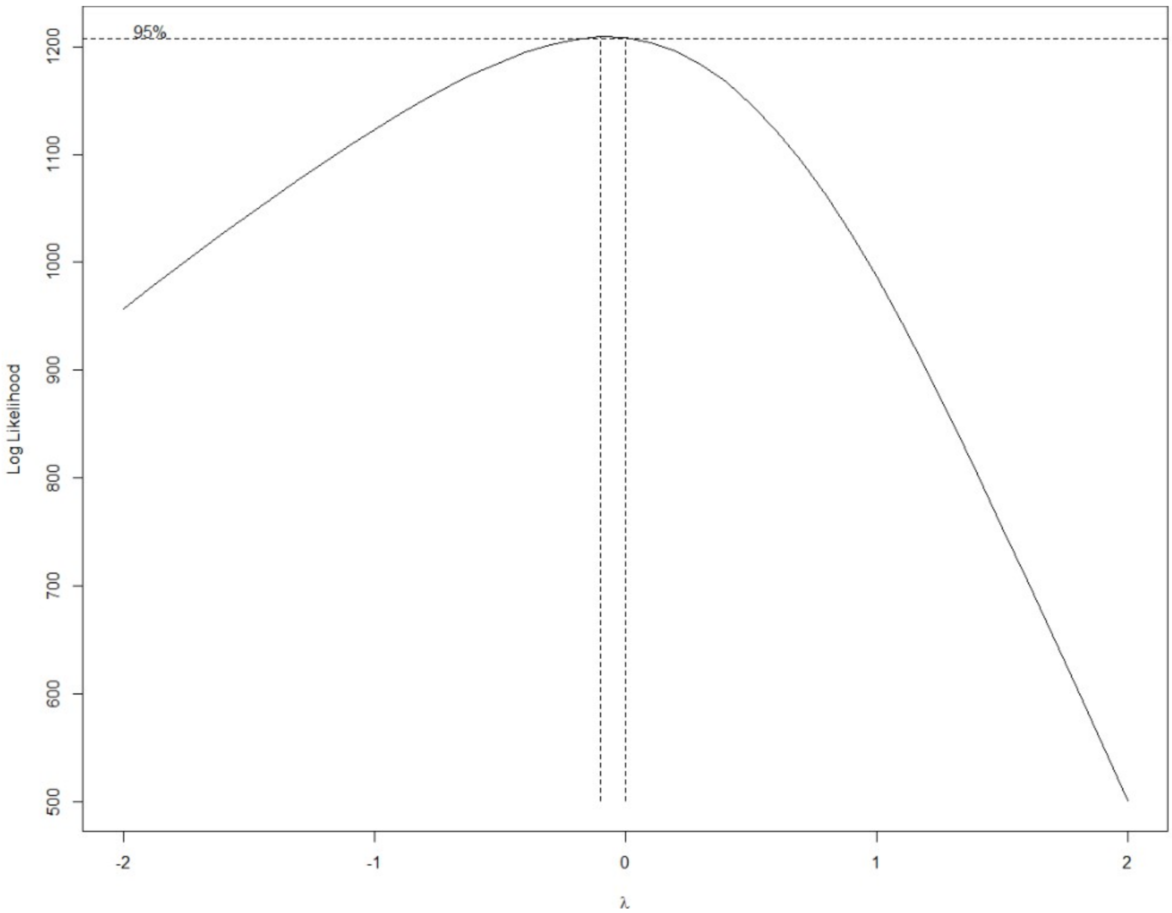
Figure 10. 10-Month Forecasts – Seasonal + Quadratic Trend Model



3.3 ARIMA Modelling

The descriptive analysis revealed strong persistence and a pronounced upward trend in the AI search interest series, indicating that the data were unlikely to satisfy the stationarity assumption required for ARIMA modelling. Consequently, a series of transformation and stationarity checks were performed before model identification.

Figure 11 presents a QQ plot of the original series. The observations deviate substantially from the reference line, particularly in the upper tail, indicating strong positive skewness and non-normal behaviour. The Shapiro–Wilk test confirmed this conclusion ($p < 0.001$), suggesting that the distribution of the raw data differs significantly from normality.



To stabilise the variance, a Box–Cox transformation was applied. The Box–Cox profile log-likelihood selected an optimal transformation parameter of $\lambda = -0.1$, with the 95% confidence interval containing values between −0.1 and 0.0. Figure 12 displays the transformed series, while Figure 13 shows the corresponding QQ plot. The transformed data exhibit improved symmetry compared with the original series, although departures from normality remain visible. The Shapiro–Wilk test statistic improved considerably after transformation, indicating that the Box–Cox transformation successfully reduced the severity of skewness.

Figure 12. Box-Cox Transformed Series

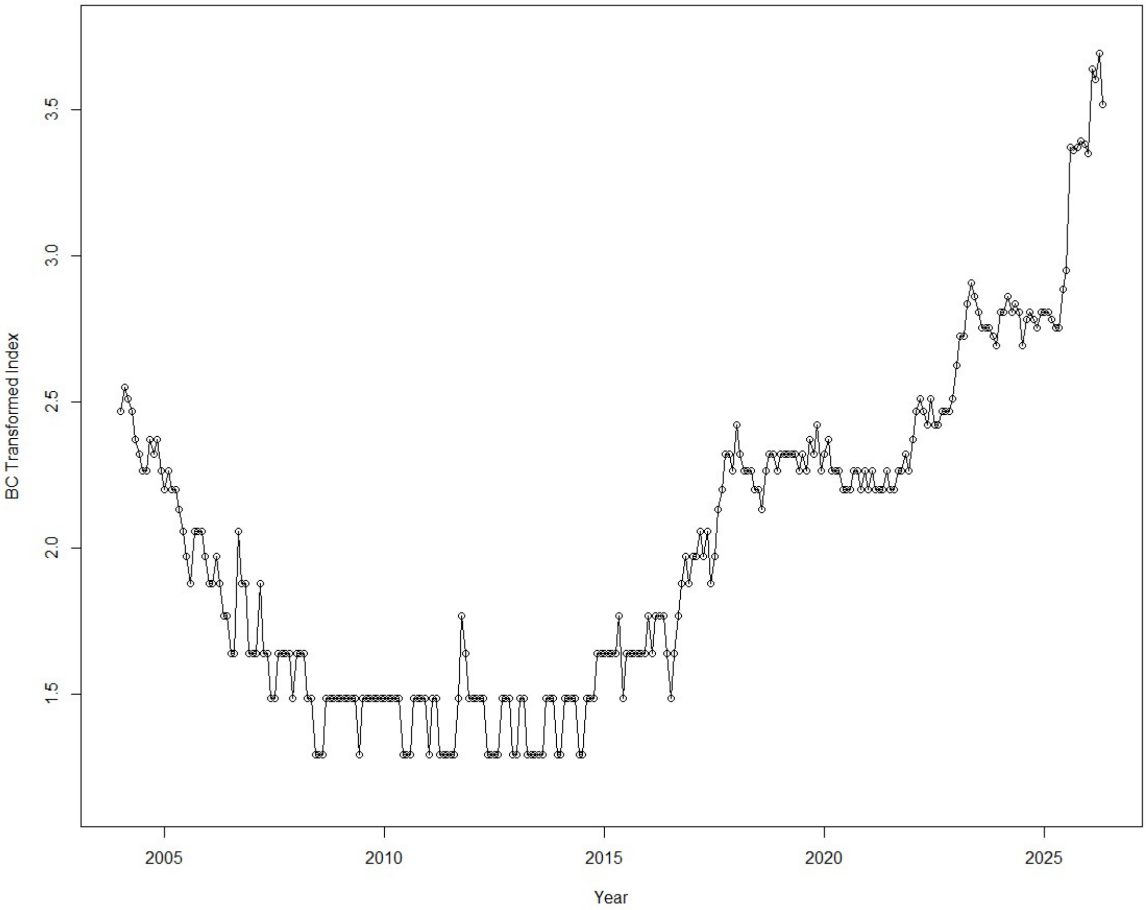
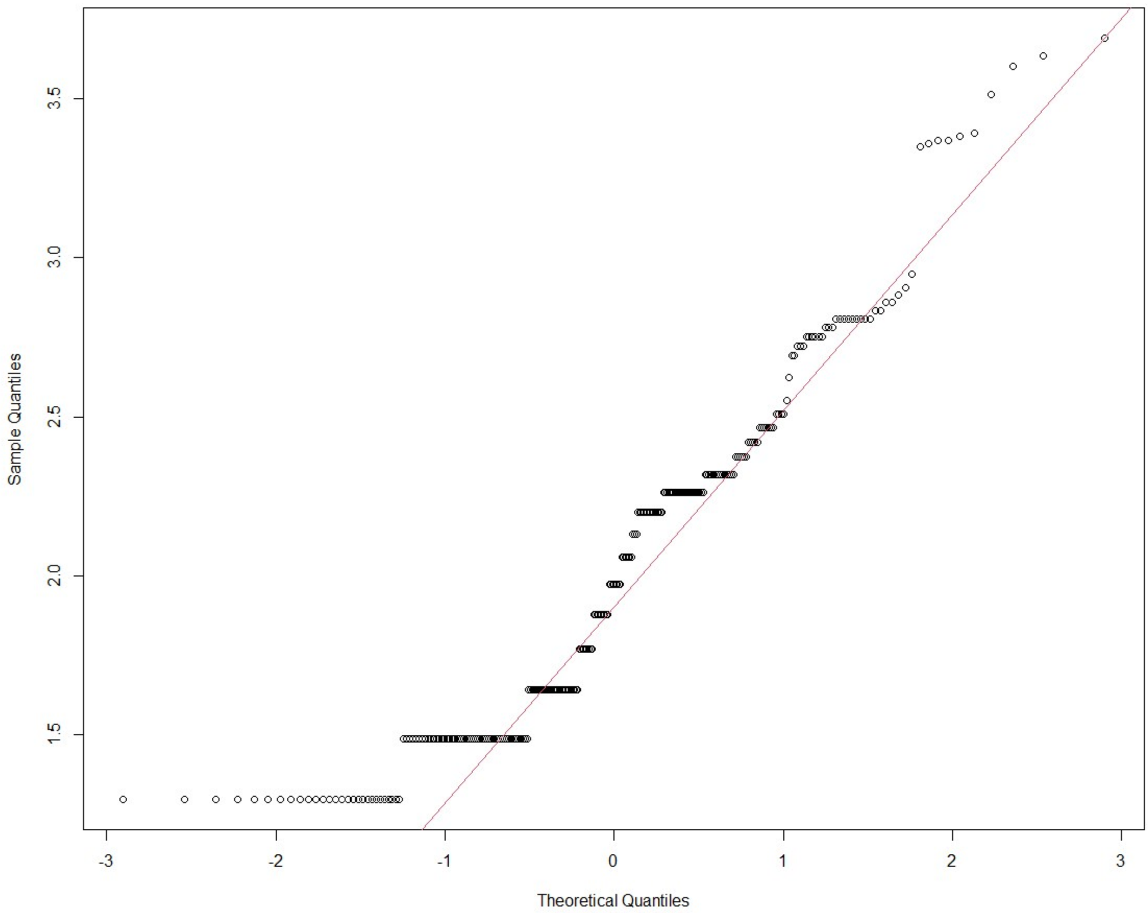


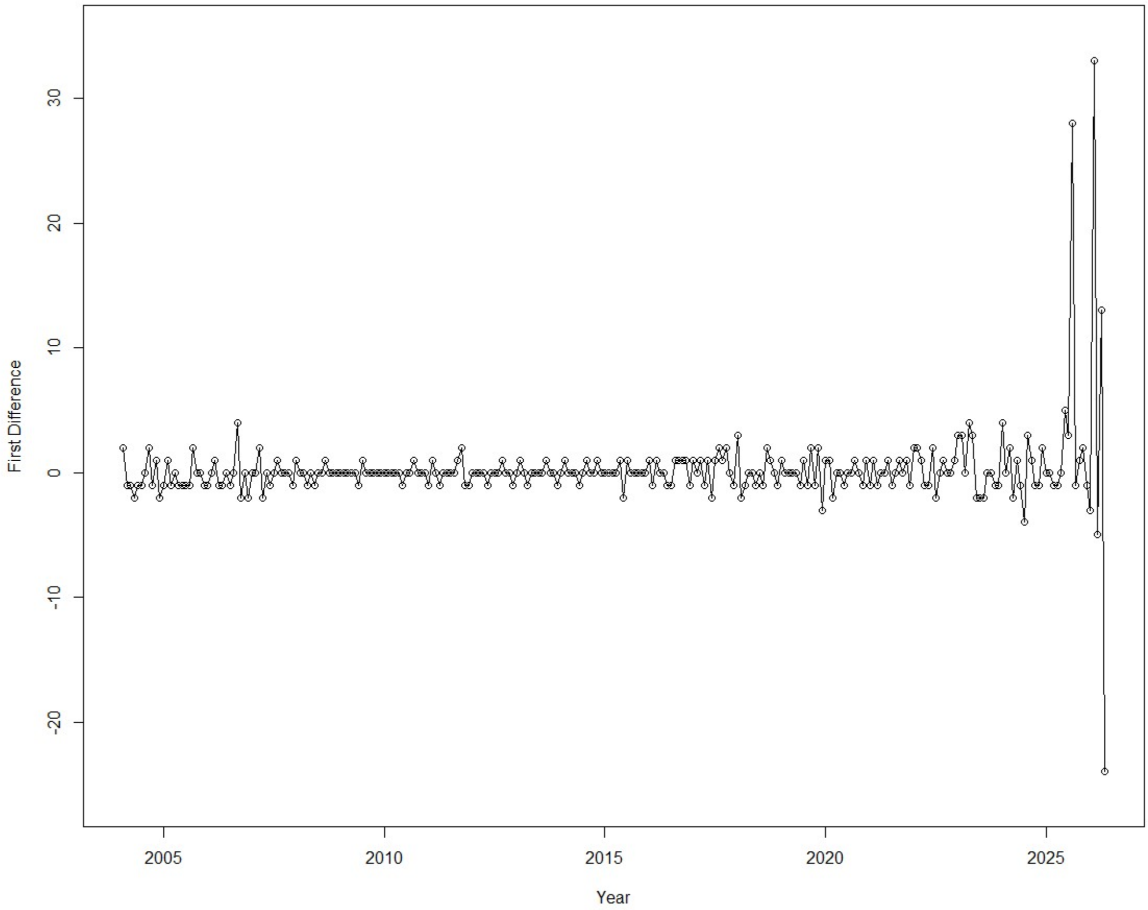
Figure 13. QQ Plot – BC Transformed Series



Stationarity was formally assessed using the Augmented Dickey–Fuller (ADF) and Phillips–Perron (PP) tests. For the original series, the ADF test produced a p-value of 0.99, while the PP test also failed to reject the null hypothesis of a unit root. These results indicate that the original AI search interest series is non-stationary and requires differencing before ARIMA modelling can proceed.

To remove the trend component, first-order differencing was applied. Figure 14 illustrates the resulting differenced series. Compared with the original data, the first-differenced series fluctuates around a relatively constant mean, suggesting that the major trend component has been successfully removed. The differenced series appears substantially more stationary and therefore suitable for ARIMA model identification.

Figure 14. First Difference of AI Search Interest



The stationarity of the differenced series was reassessed using formal statistical tests. The ADF test rejected the null hypothesis of a unit root, providing evidence that the differenced series is stationary. The PP test produced a consistent conclusion. These results indicate that first-order differencing was sufficient to remove the non-stationary trend component and satisfy the stationarity requirement for ARIMA modelling.

The autocorrelation structure of the stationary series was then examined using the ACF and PACF plots shown in Figures 15 and 16. The ACF displays a significant negative spike at lag 1 followed by a rapid decay toward zero, while the PACF exhibits only a small number of significant lags. This pattern suggests that the differenced series may be adequately represented by a low-order ARIMA model with one or more moving-average components. However, the identification was not completely straightforward, and additional model selection tools were used to determine an appropriate model structure.

Figure 15. ACF – First Difference

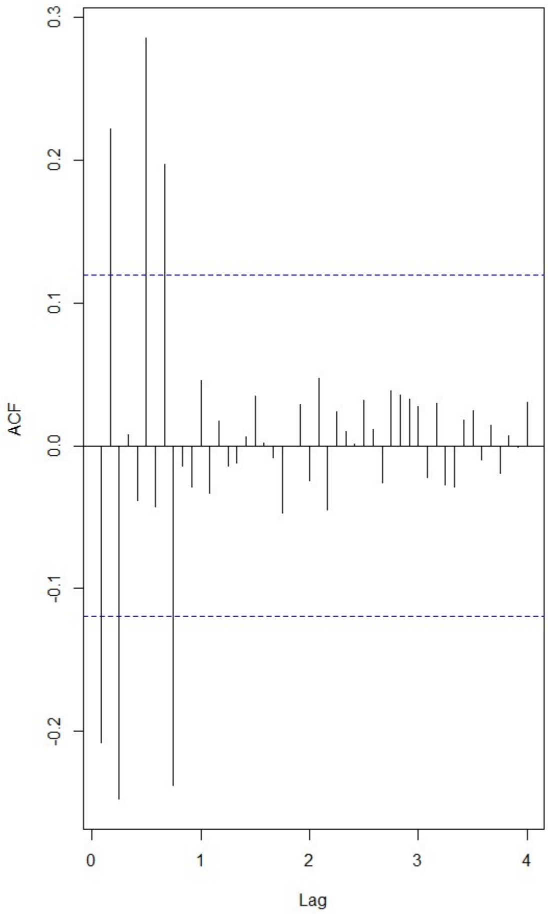
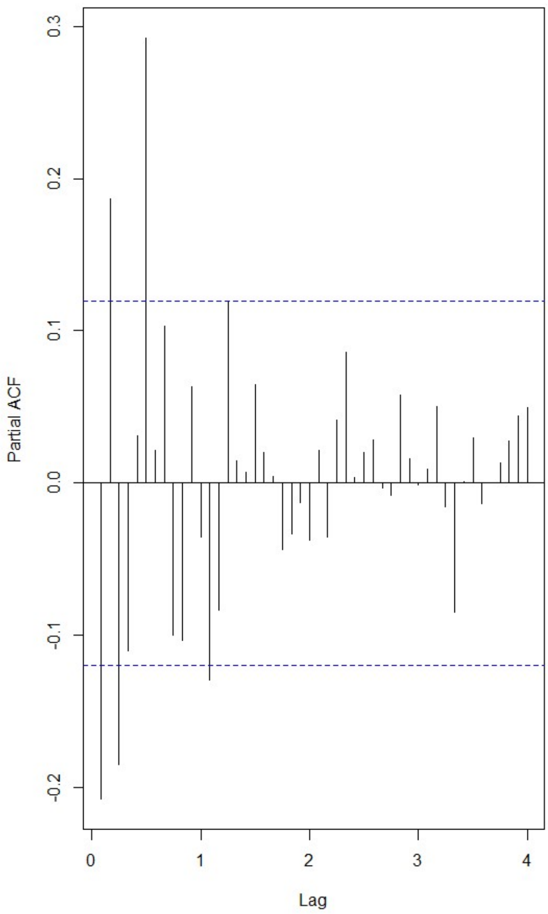
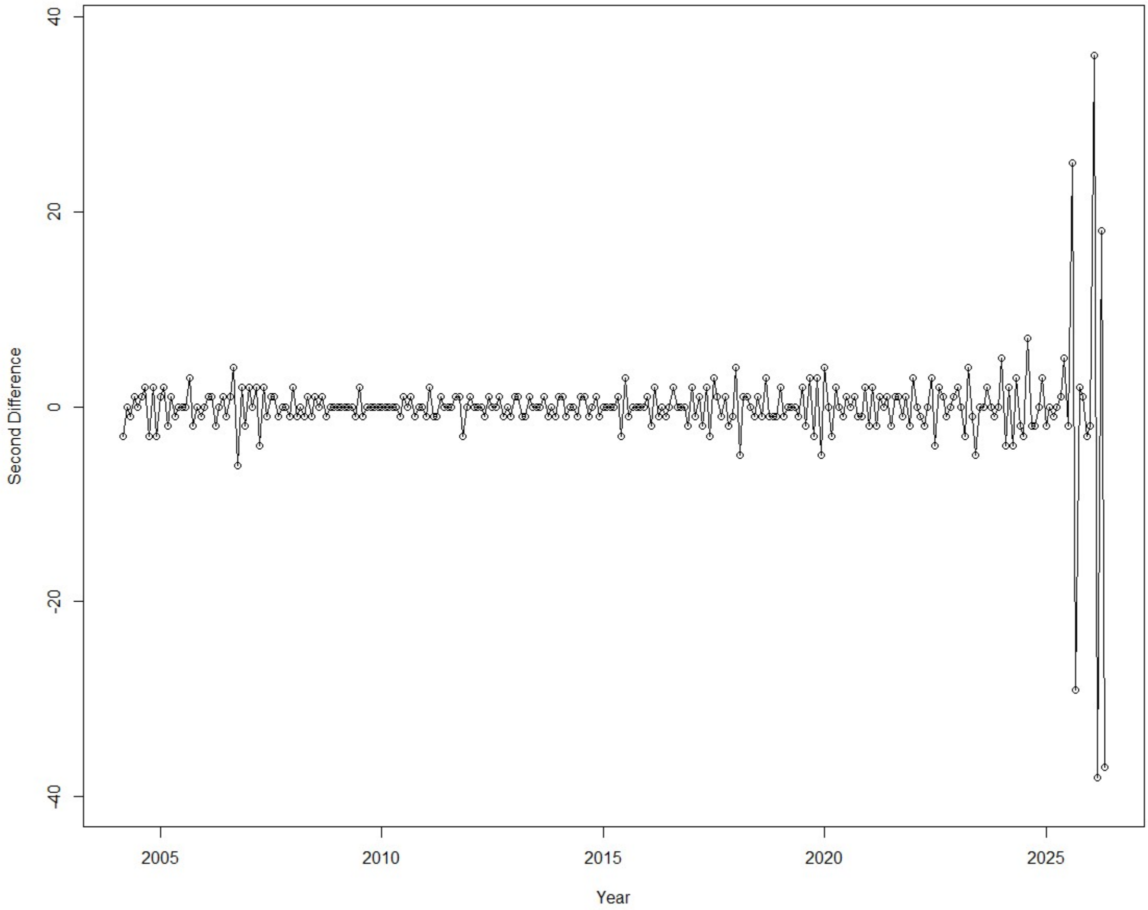


Figure 16. PACF – First Difference



The autocorrelation structure of the differenced series was further examined using the second-differenced series shown in Figure 17 together with the ACF and PACF plots presented in Figures 18 and 19. The ACF exhibits a rapid decay after the first few lags, while the PACF contains only a small number of significant spikes. These patterns suggest that a low-order ARIMA specification is appropriate for modelling the stationary series. Based on these diagnostics, a set of candidate models was selected for estimation, including ARIMA(0,1,1), ARIMA(1,1,1), ARIMA(2,1,1), ARIMA(0,1,2), ARIMA(1,1,2), and ARIMA(2,1,2). These models were fitted using both maximum likelihood (ML) and conditional sum-of-squares (CSS) estimation methods.

Figure 17. Second Difference of AI Search Interest



Residual diagnostics were performed to assess model adequacy. The residual series fluctuates randomly around zero, and the residual autocorrelation function showed no substantial remaining autocorrelation. The Ljung–Box test produced a p-value of 0.09, indicating that the residuals do not significantly differ from white noise. Furthermore, the Shapiro–Wilk test yielded a p-value of 0.18, suggesting no strong evidence against normality. Collectively, these results indicate that the ARIMA(0,1,1) model provides an adequate representation of the non-seasonal dynamics in the AI search interest series.

Figure 18. ACF – Second Difference

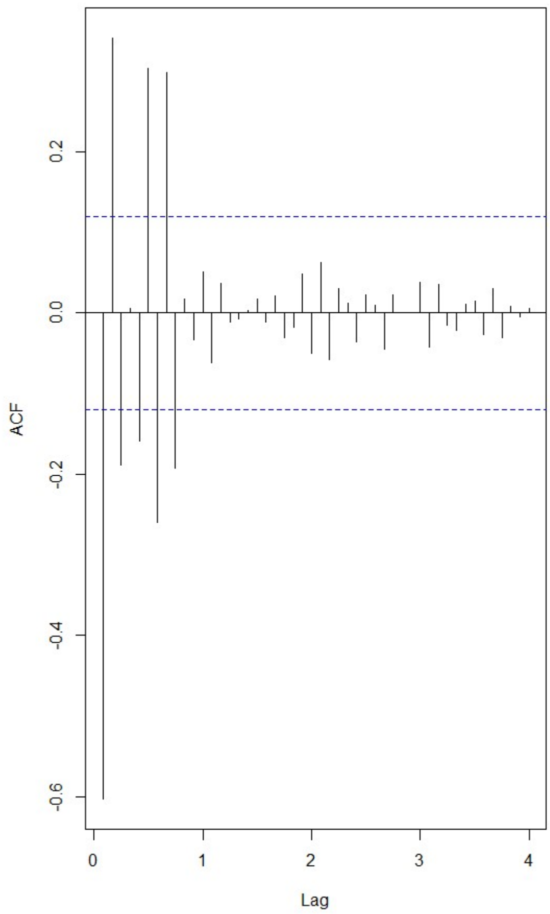
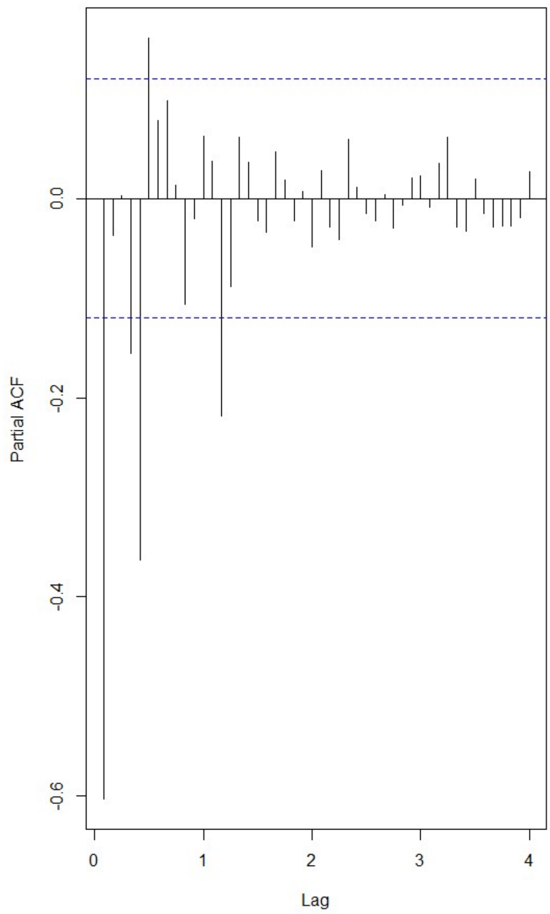
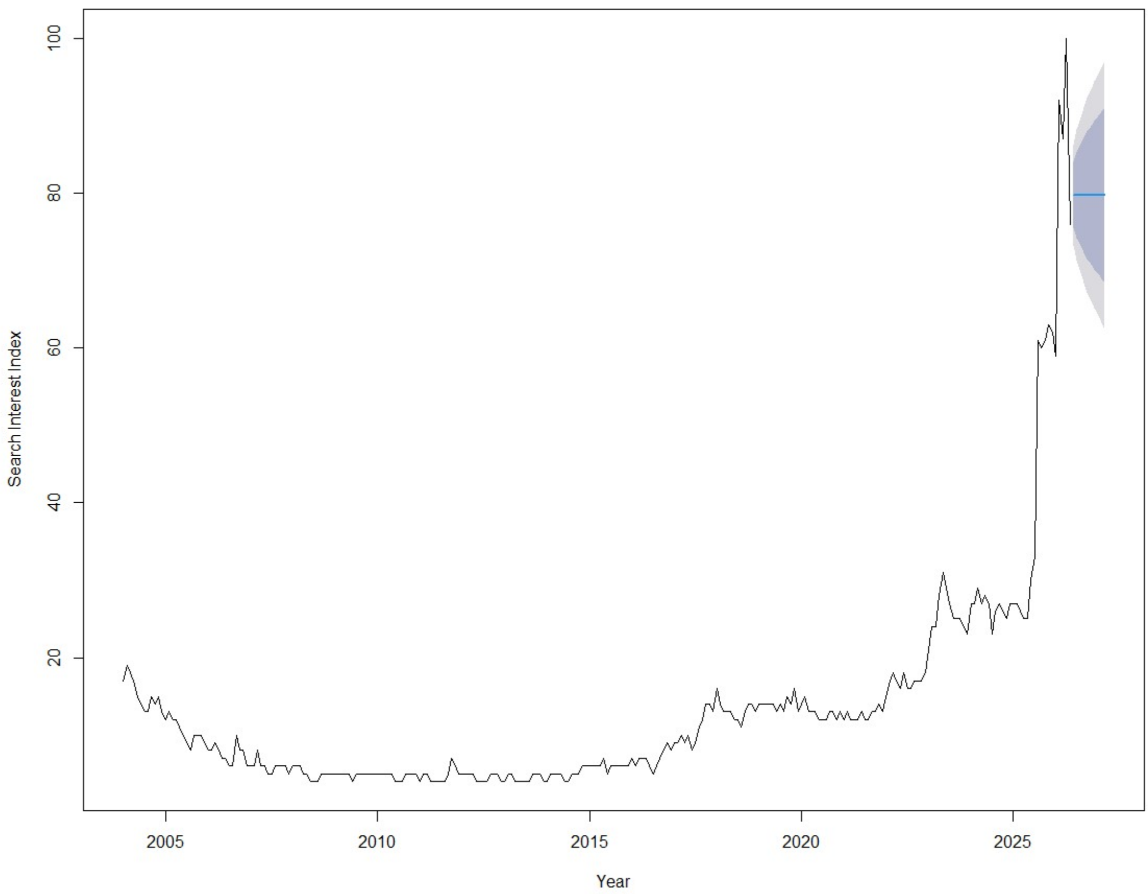


Figure 19. PACF – Second Difference



Finally, the selected ARIMA model was used to generate forecasts for the next ten months. The resulting forecasts are presented in Figure 20. The model predicts that global interest in Artificial Intelligence will remain at historically high levels throughout the forecast period. Although the point forecasts continue to increase gradually, the widening prediction intervals indicate growing uncertainty as the forecasting horizon extends further into the future. The ARIMA forecasts provide a useful benchmark against which the seasonal and volatility-based models are compared in subsequent sections.

Figure 20. 10-Month Forecasts – ARIMA Model



3.4 SARIMA Modelling

Although the ARIMA model adequately captured the non-seasonal structure of the series, the monthly nature of the data suggested that seasonal dependence may also be present. Therefore, Seasonal ARIMA (SARIMA) models were investigated to account for both non-seasonal and seasonal patterns.

The seasonal autocorrelation and partial autocorrelation functions are presented in Figures 21 and 22. Significant spikes at seasonal lags, particularly around lag 12, suggest the presence of seasonal dependence and support the use of SARIMA modelling. Figure 23 shows the series after seasonal differencing, which fluctuates around a more stable mean and exhibits reduced seasonal structure compared with the original series.

Figure 21. Seasonal ACF – AI Search Interest

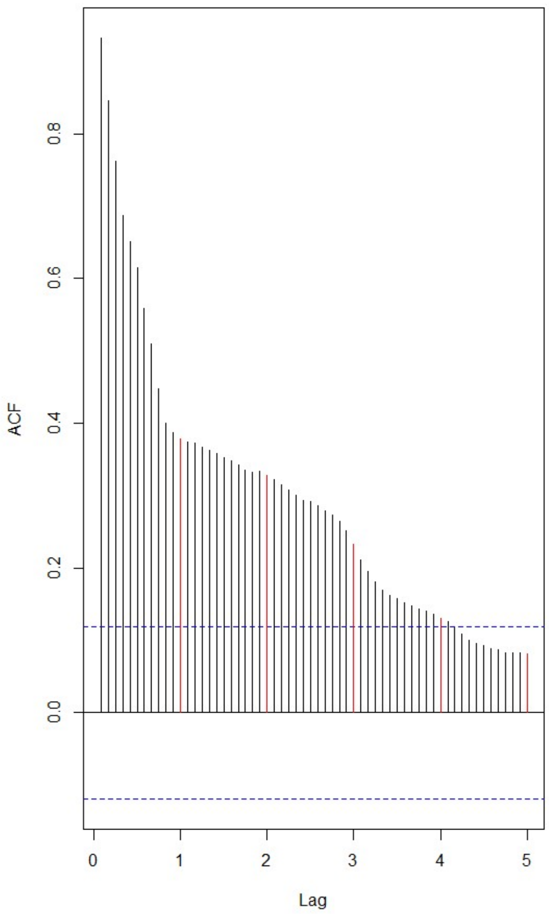
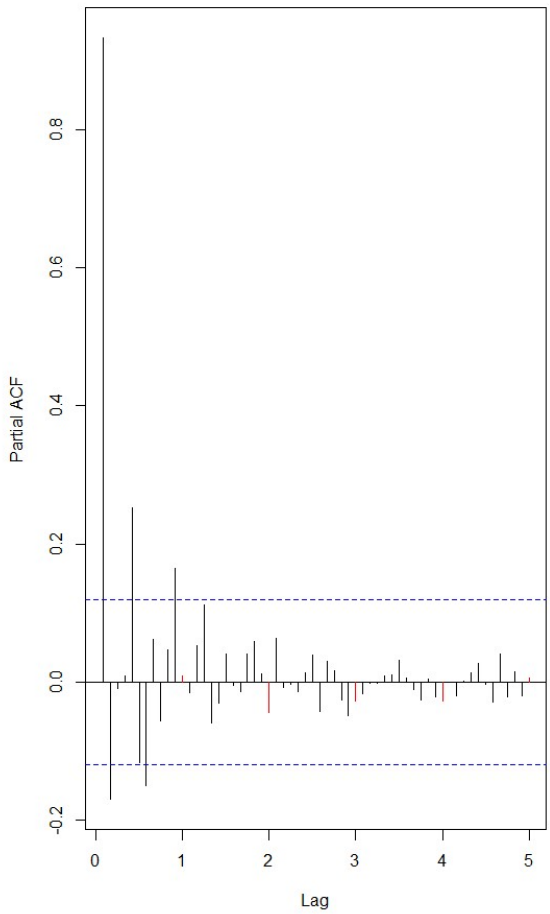


Figure 22. Seasonal PACF – AI Search Interest



Based on the seasonal ACF and PACF patterns, several candidate SARIMA models were considered. The candidate models included SARIMA(0,1,1)×$(0,1,1)_{12}$, SARIMA(1,1,1)×$(0,1,1)_{12}$, SARIMA(0,1,2)×$(0,1,1)_{12}$, and SARIMA(1,1,2)×$(0,1,1)_{12}$.

Figure 23. Residuals after Seasonal Difference

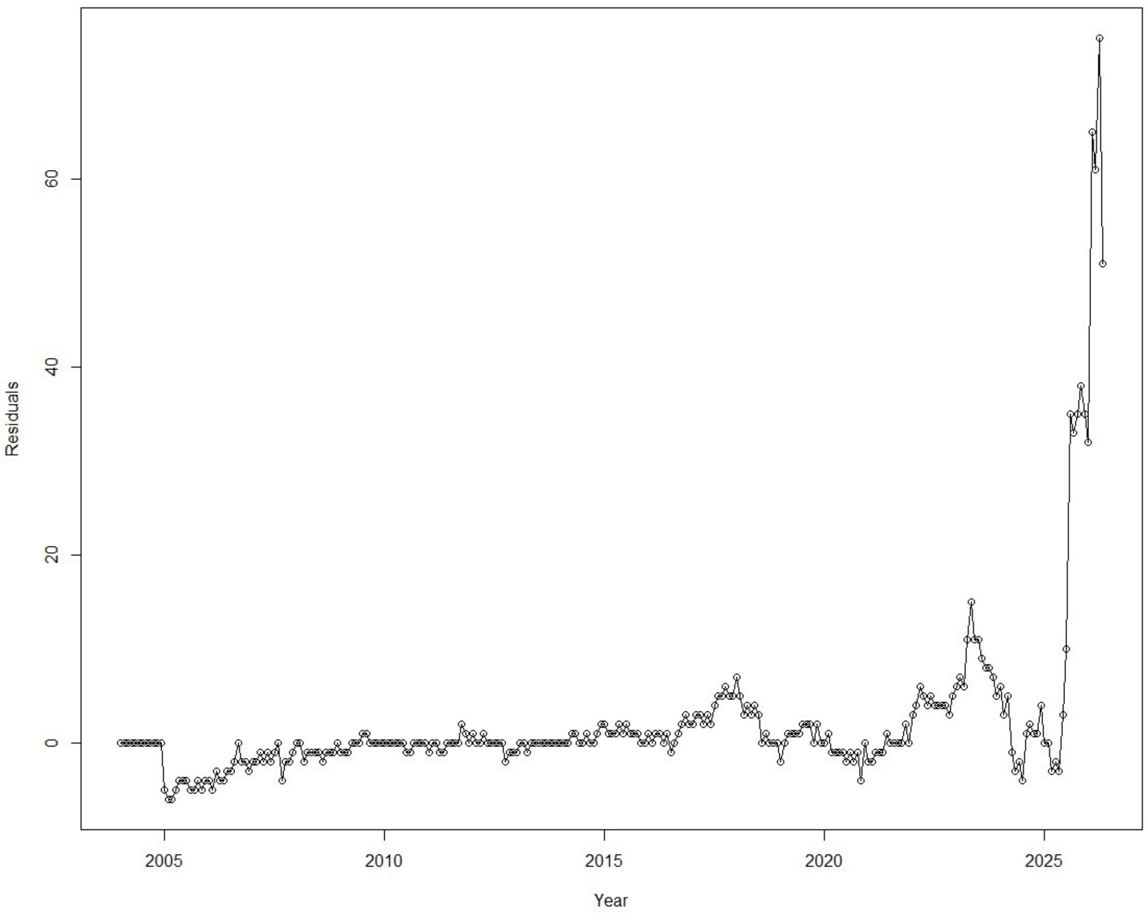


Figure 24. Residuals after SARMA(0,1,1)_12

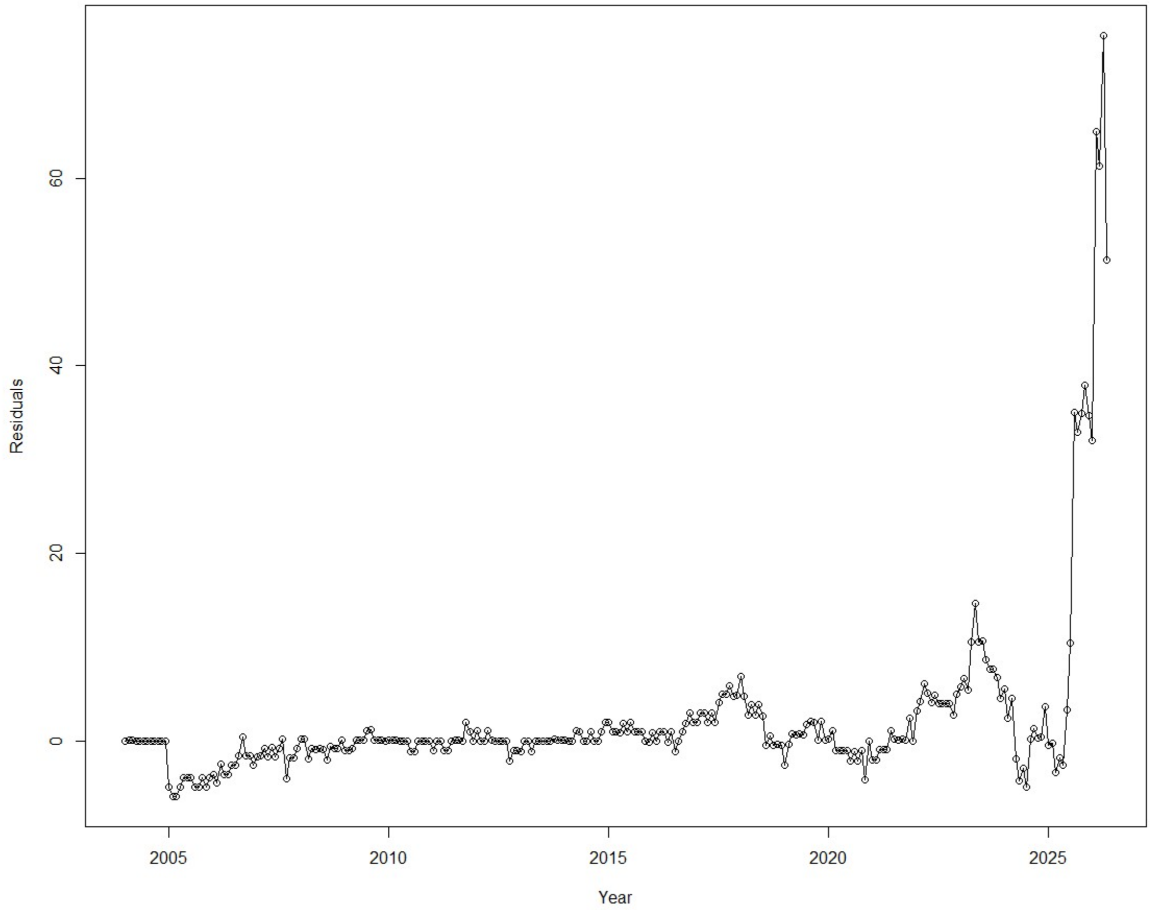


Table 3 summarises the AIC and BIC values for the fitted SARIMA models. Among the candidate specifications, SARIMA(0,1,1)×(0,1,1)$_{12}$ achieved the lowest AIC (1798.62) and BIC (1811.86), indicating the most parsimonious and best-fitting model.

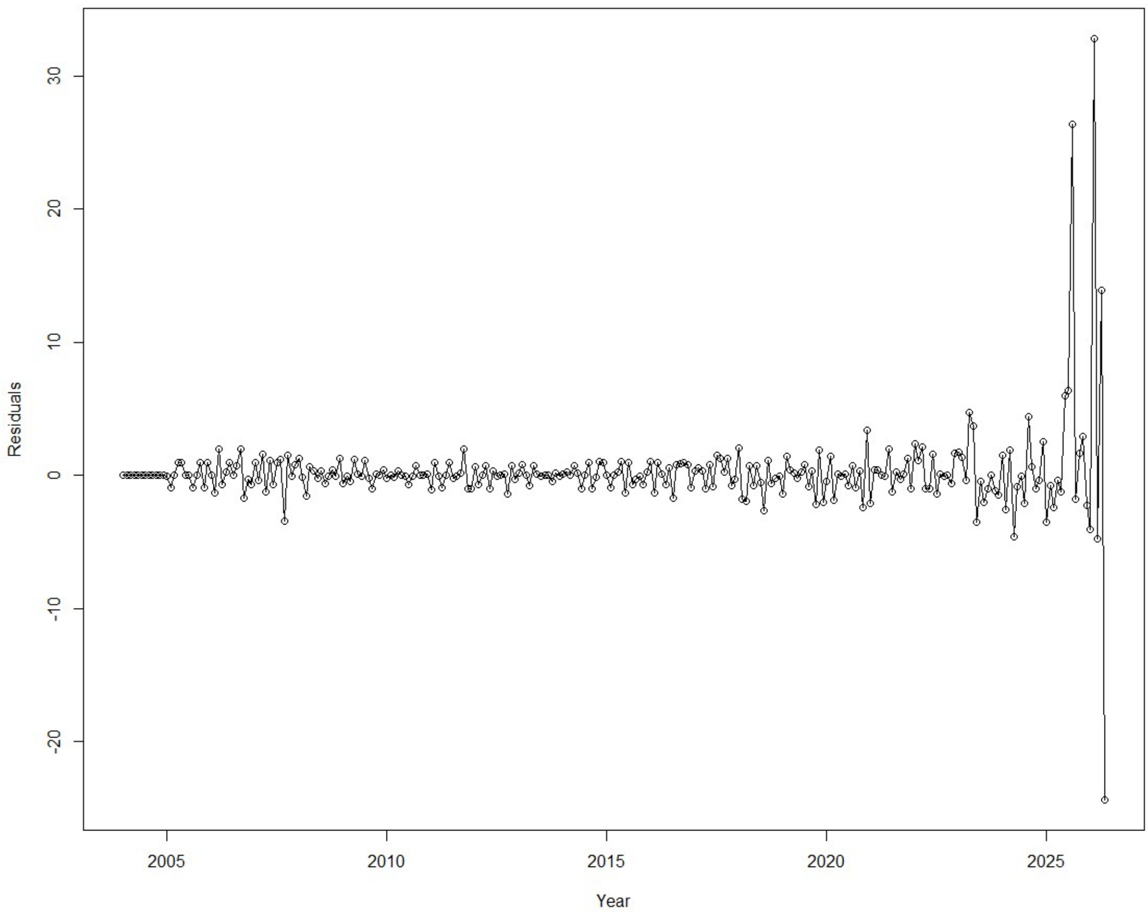
Table 3. Comparison of candidate SARIMA models

Model	AIC	BIC
SARIMA(0,1,1)(0,1,1)$_{12}$	1798.62	1811.86
SARIMA(1,1,1)(0,1,1)$_{12}$	1800.58	1817.08
SARIMA(0,1,2)(0,1,1)$_{12}$	1801.93	1818.43
SARIMA(1,1,2)(0,1,1)$_{12}$	1802.57	1822.07

The estimated parameters were statistically significant. The non-seasonal moving-average coefficient was estimated at −0.184, while the seasonal moving-average coefficient was estimated at −0.483. Both coefficients were significantly different from zero, indicating that recent shocks and seasonal shocks contribute meaningfully to future observations.

Residual diagnostics were performed to evaluate model adequacy. The residual series exhibited no obvious systematic patterns, and the residual autocorrelation function contained no substantial significant spikes. The Ljung–Box test produced a p-value of 0.14, suggesting that the residuals are consistent with white noise. In addition, the McLeod–Li test did not indicate remaining ARCH effects. These results suggest that the SARIMA$(0,1,1)\times(0,1,1)_{12}$ model successfully captured the primary temporal structure of the series.

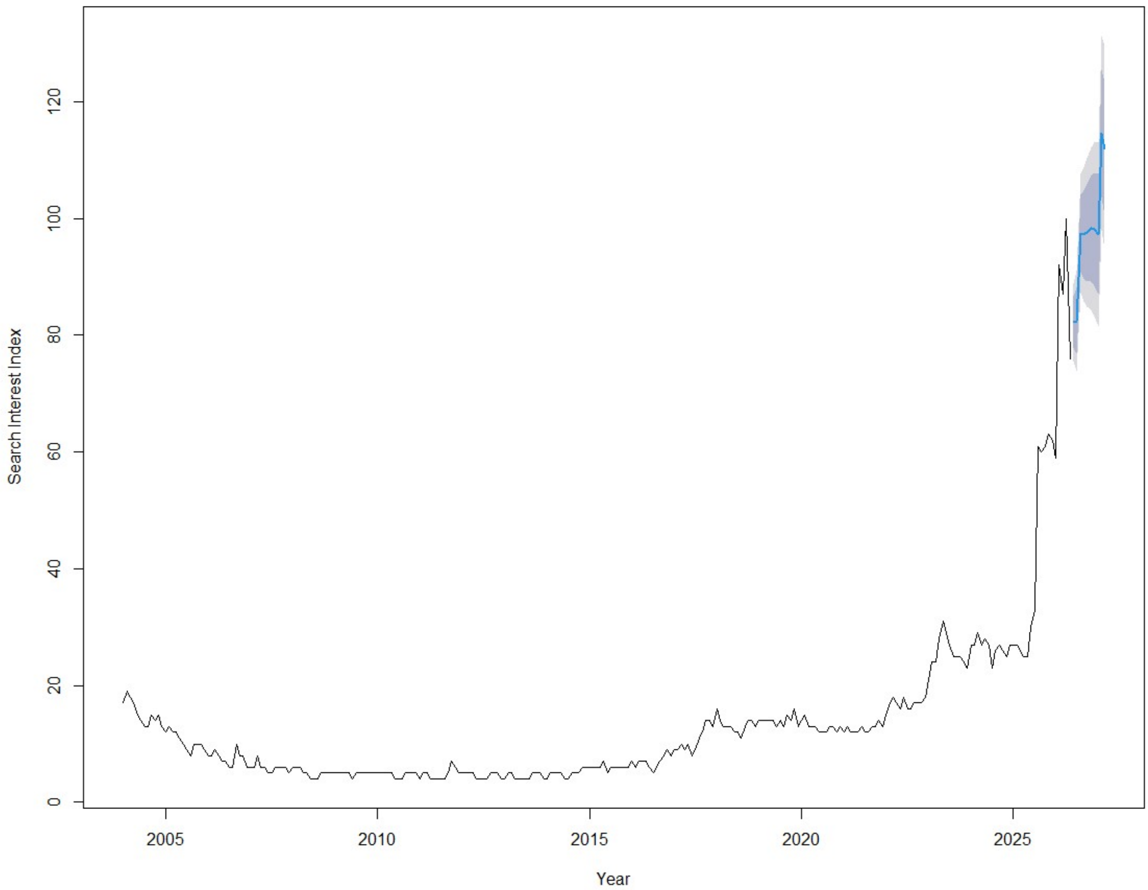
Figure 25. Residuals after d=1, D=1



The forecasting performance of the selected SARIMA model was evaluated against the competing non-seasonal ARIMA model. The SARIMA$(0,1,1)\times(0,1,1)_{12}$ model achieved an RMSE of 3.269 and a MAE of 1.184, marginally outperforming the ARIMA(0,1,1) model on both measures. Although the ARIMA model achieved a slightly lower MAPE of 8.225% compared with 8.692% for the SARIMA model, the SARIMA specification was preferred overall due to its lower AIC and BIC and its explicit modelling of seasonal dependence. These results indicate that incorporating seasonal structure improves average forecast error on the primary accuracy measures.

Figure 26 presents the 10-month forecasts generated from the SARIMA$(0,1,1)\times(0,1,1)_{12}$ model. The forecasts indicate that global interest in Artificial Intelligence is expected to remain at historically high levels throughout the forecast horizon. Point forecasts for June and July 2026 are approximately 82, before rising to approximately 97–98 from August 2026 onwards, with February and March 2027 exceeding 100 on the index scale. Since the Google Trends index is bounded at 100, values above this threshold should be interpreted as indicating that AI search interest is expected to remain at or near the maximum observed level. The prediction intervals widen gradually over time, reflecting increasing forecast uncertainty at longer horizons.

Figure 26. 10-Month Forecasts – SARIMA Model

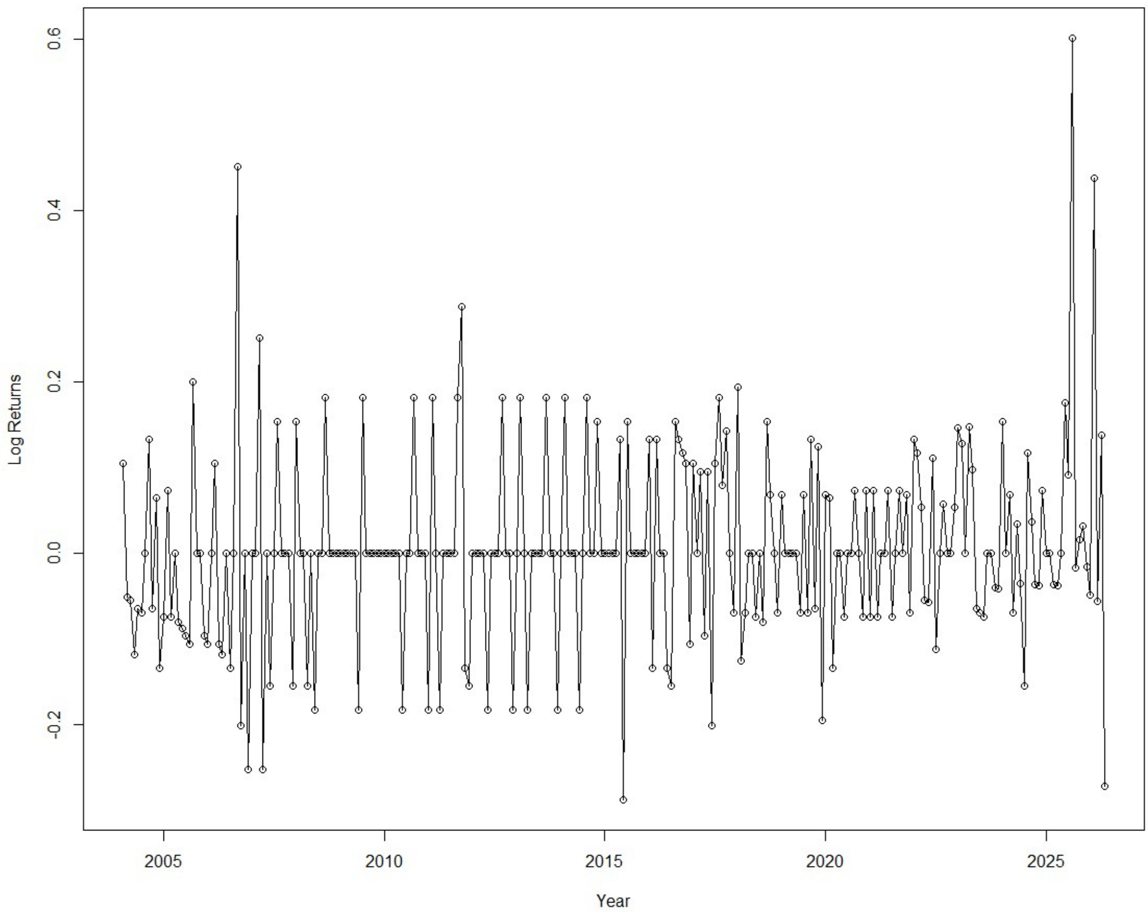


3.5 GARCH Modelling

While ARIMA and SARIMA models focus on modelling the conditional mean of a time series, they assume that the variance remains constant over time. Inspection of the AI search interest series suggested periods of increased volatility, particularly following the rapid growth in public interest after 2022. Therefore, GARCH modelling was considered to examine whether the variability of the series changes over time and whether modelling this volatility improves forecasting performance.

To satisfy the assumptions of GARCH modelling, the analysis was performed on the continuously compounded return series derived from the AI search interest index. Figure 27 presents the transformed return series. Although the mean appears relatively stable, periods of high and low volatility are visible, suggesting possible volatility clustering. Such behaviour is commonly observed in financial and internet search activity data and provides motivation for applying GARCH models.

Figure 27. Returns of AI Search Interest (diff of log)



The autocorrelation structure of the return series was examined using the ACF and PACF plots shown in Figures 28 and 29. The correlations were generally weak, indicating that little serial dependence remained in the mean process. Consequently, attention was focused on modelling the variance dynamics through an ARMA–GARCH framework.

Figure 28. McLeod-Li Test – AI Returns

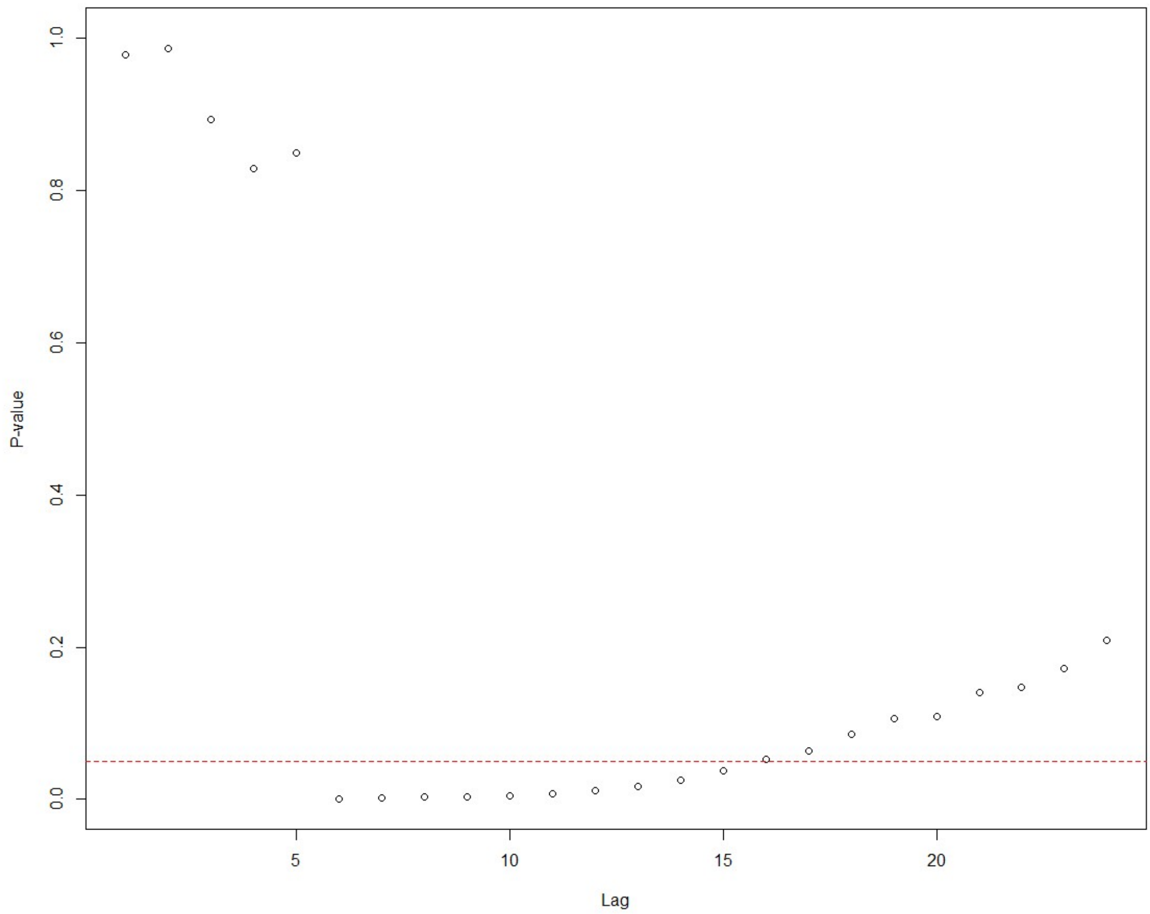


Figure 29. ACF of Returns

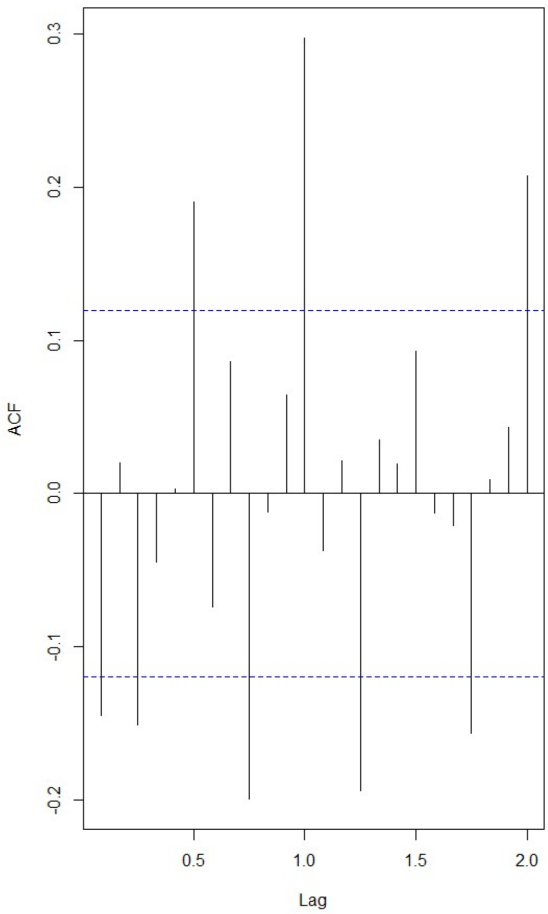
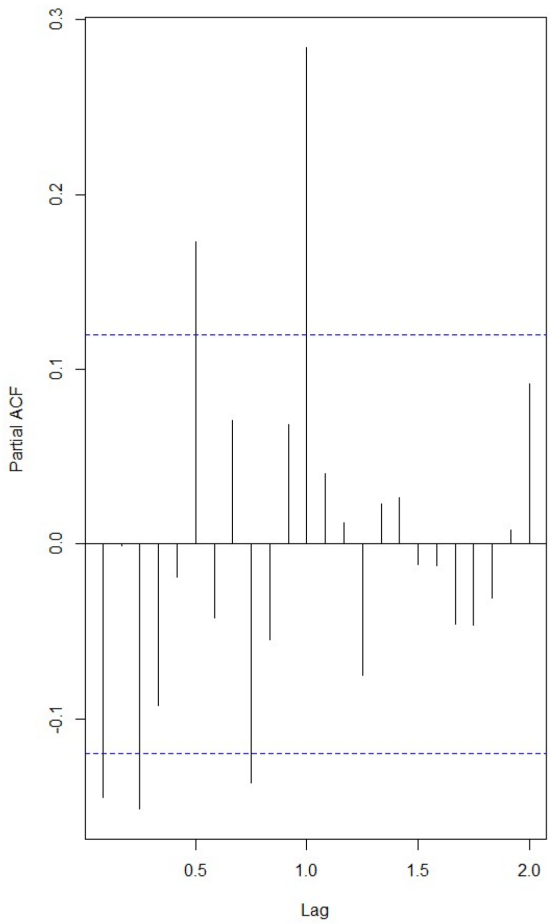


Figure 30. PACF of Returns



Model identification was supported using the BIC subset selection procedure shown in Figure 30. Several candidate mean models were considered, with ARMA(0,1) emerging as the most parsimonious specification. This model was subsequently combined with alternative GARCH structures, including GARCH(0,1), GARCH(1,1), and GARCH(1,2). Figure 31 presents the ACF of the absolute residuals, while Figure 32 shows the corresponding PACF. The persistence of correlations in the absolute residuals provided evidence of conditional heteroscedasticity and justified the use of GARCH models.

Figure 31. ACF – |Residuals|

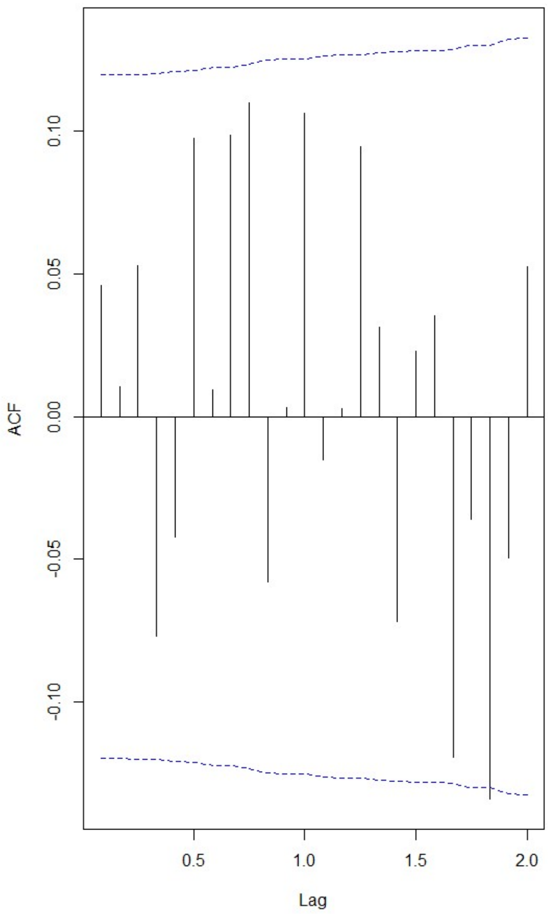


Figure 32. PACF – |Residuals|

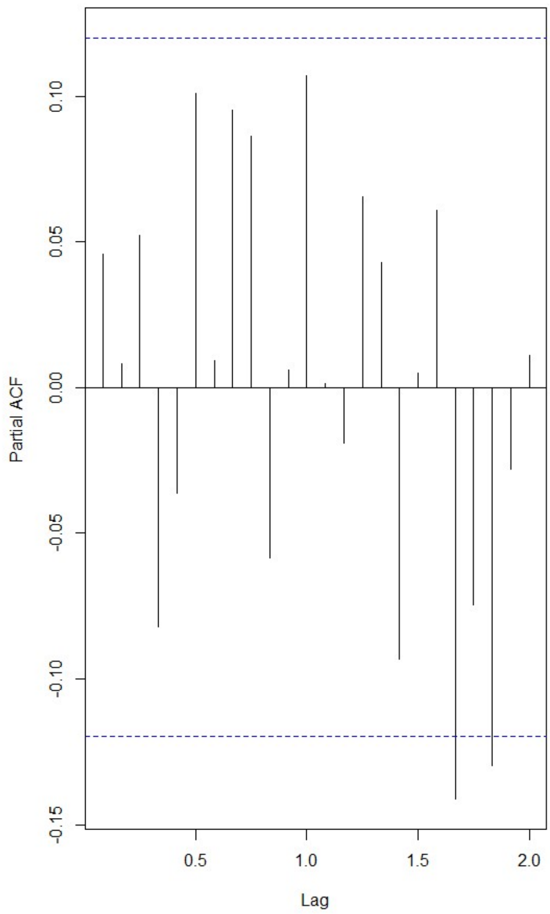


Table 4 summarises the fitted ARMA–GARCH models. Among the competing specifications, the ARMA(0,1)+GARCH(1,1) model achieved the lowest information criteria values and was therefore selected as the preferred volatility model. The estimated parameters satisfied the stationarity conditions and indicated that volatility shocks persist over time, although their effects gradually diminish.

Table 4. Comparison of fitted ARMA-GARCH models

Model	AIC	BIC
ARMA(0,1)+GARCH(0,1)	-3.812	-3.774
ARMA(0,1)+GARCH(1,1)	-3.841	-3.798
ARMA(0,1)+GARCH(1,2)	-3.829	-3.781

Figure 33 displays the standardized residuals from the fitted ARMA(0,1)+GARCH(1,1) model. The residuals fluctuate randomly around zero with no obvious systematic patterns, suggesting that the model adequately captures the conditional mean and variance dynamics of the series.

Furthermore, no extended periods of unusually high or low volatility remain visible after standardization.

Figure 34 presents the PACF of the residuals from the fitted ARMA(0,1)+GARCH(1,1) model. The majority of partial autocorrelations lie within the significance bounds, with no substantial spikes observed at higher lags. This suggests that little serial dependence remains in the residual series after model fitting. Together with the ACF and PACF analyses presented in Figures 31–34, these results indicate that the GARCH specification successfully captured the volatility clustering and dependence structure present in the return series, supporting the adequacy of the fitted model.

Figure 33. ACF – Residuals²

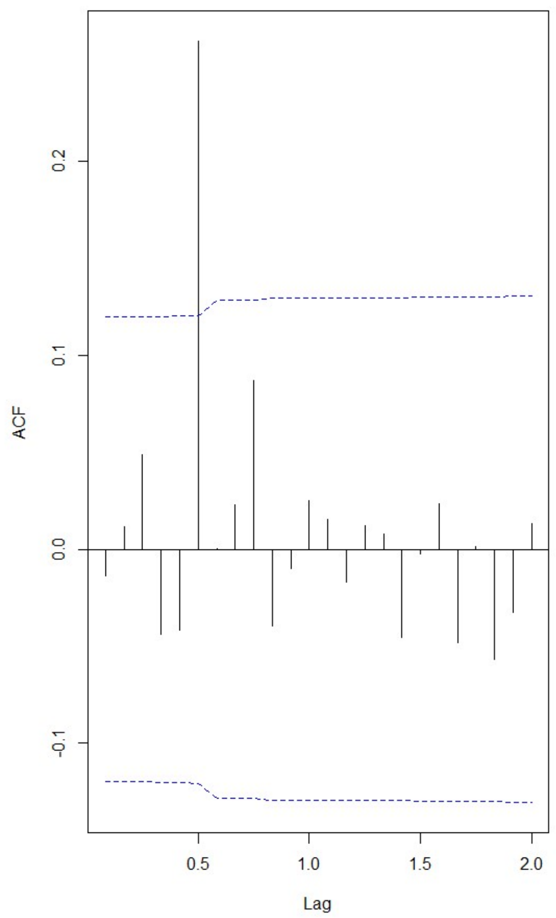
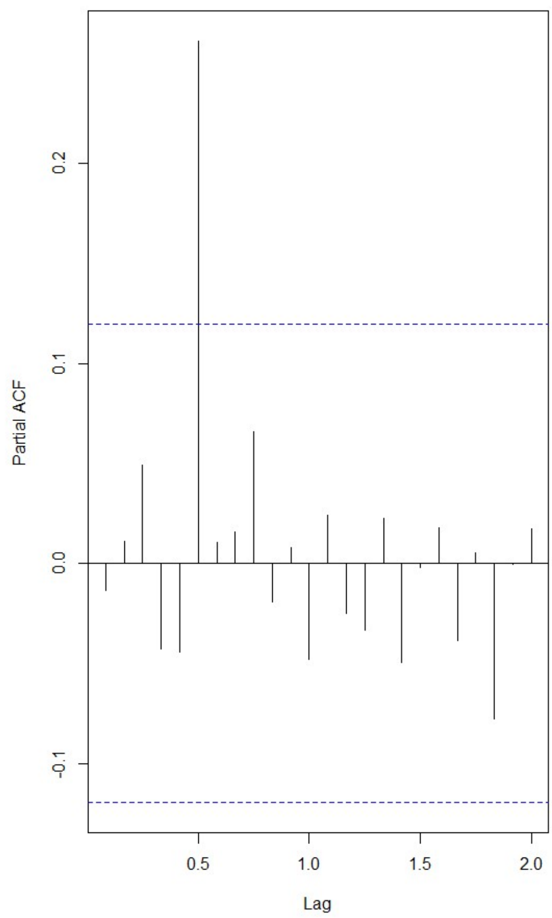


Figure 34. PACF – Residuals²



Additional diagnostic tests supported these graphical findings. The McLeod–Li test did not detect significant remaining ARCH effects, indicating that the fitted model adequately explains the conditional heteroscedasticity present in the data. Overall, the ARMA(0,1)+GARCH(1,1) model provides an appropriate representation of the volatility behaviour of AI search interest.

Although the GARCH model effectively captured changes in volatility, its primary contribution was to modelling uncertainty rather than improving point forecasts of the original search interest series. Consequently, while the ARMA(0,1)+GARCH(1,1) model provides valuable insights into volatility dynamics, the SARIMA model remained the preferred forecasting model for the original monthly AI search interest data.

3.6 AIC-BIC Comparison Table

The main candidate models were compared using information criteria, forecast accuracy measures, and residual diagnostic results. AIC and BIC were used because they balance model goodness-of-fit with model complexity. Lower AIC and BIC values generally indicate a better model, although the GARCH model values should not be directly compared with the ARIMA and SARIMA values because the GARCH models were fitted to the return series rather than the original search-interest index.

Table 5. Summary comparison of selected candidate models

Model family	Selected model	AIC	BIC	Main purpose
Trend regression	Seasonal + Quadratic Trend Model	1868.14	-	Captures long-term nonlinear trend and seasonal effects
ARIMA	ARIMA(0,1,1)	1821.00	1828.00	Captures non-seasonal autocorrelation after differencing
SARIMA	$SARIMA(0,1,1)x(0,1,1)_{12}$	1798.62	1811.86	Captures both non-seasonal and seasonal dependence
ARMA-GARCH	ARMA(0,1)+GARCH(1,1)	-3.841	-3.798	Captures changing volatility in the return series

The comparison indicates that the $SARIMA(0,1,1)\times(0,1,1)_{12}$ model achieved the strongest performance among the forecasting models fitted to the original monthly search-interest series. It produced lower AIC and BIC values than the selected ARIMA model, showing that incorporating seasonal dependence improved model fit. The Seasonal + Quadratic Trend Model was useful as a baseline model, but it did not model autocorrelation in the residuals as directly as the ARIMA-family models. The GARCH model was valuable for analysing volatility, but it was not selected as the final forecasting model for the original Google Trends index because it was fitted to returns rather than the original level series.

3.7 Forecasting Table

The final selected forecasting model was $SARIMA(0,1,1)\times(0,1,1)_{12}$. This model was used to generate 10-month ahead forecasts from June 2026 to March 2027. The forecast table includes the point forecasts and the 80% and 95% prediction intervals.

Table 6. Ten-month forecasts from the $SARIMA(0,1,1)x(0,1,1)_{12}$ model

Month	Point Forecast	80% Lower	80% Upper	95% Lower	95% Upper
Jun 2026	82.27	77.94	86.60	75.64	88.90
Jul 2026	82.44	76.85	88.04	73.89	91.00

Aug 2026	97.43	90.81	104.04	87.31	107.55
Sep 2026	97.28	89.78	104.78	85.81	108.75
Oct 2026	97.55	89.25	105.84	84.86	110.23
Nov 2026	98.24	89.23	107.26	84.45	112.03
Dec 2026	98.12	88.44	107.80	83.31	112.93
Jan 2027	97.28	86.98	107.59	81.52	113.05
Feb 2027	114.60	103.70	125.50	97.93	131.27
Mar 2027	112.00	100.55	123.46	94.48	129.53

The forecasts suggest that worldwide search interest in Artificial Intelligence is expected to remain very high over the forecast period. Forecasts for June and July 2026 are around 82, while forecasts from August 2026 onward remain close to or above the upper range of the Google Trends index. Since Google Trends is bounded between 0 and 100, point forecasts above 100 should not be interpreted literally. Instead, they indicate that the model expects AI search interest to remain at or near the maximum observed level.

3.8 Model Comparison

The fitted models were compared using model fit, forecast accuracy, residual behaviour, and interpretability. The trend regression models were useful for understanding the overall long-term movement in AI search interest. The linear model was too simple because it could not capture the rapid acceleration in interest after 2022. The quadratic and seasonal quadratic models improved the fitted trend, but these models did not fully account for the autocorrelation structure of the time series.

The ARIMA(0,1,1) model provided a simple and interpretable non-seasonal time series model. It captured the main non-seasonal dependence after first differencing and produced reasonable residual diagnostics. However, because the dataset is monthly, it was important to consider seasonal dependence as well.

The SARIMA(0,1,1)×(0,1,1)$_{12}$ model improved on the ARIMA model by including a seasonal moving-average component. This was appropriate because the ACF and PACF plots showed evidence of seasonal behaviour at monthly lags. The SARIMA model also achieved lower AIC and BIC values than the selected ARIMA model, indicating better overall fit after accounting for model complexity.

Table 7. Forecast accuracy comparison for selected ARIMA and SARIMA models

Model	RMSE	MAE	MAPE
ARIMA(0,1,1)	3.299	1.192	8.225
SARIMA(0,1,1)x(0,1,1)$_{12}$	3.269	1.184	8.692

The SARIMA model had a slightly lower RMSE and MAE than the selected ARIMA model, showing a small improvement in average forecast error. Although the ARIMA model had a slightly lower MAPE, the SARIMA model was preferred because it incorporated the monthly seasonal structure of the data and produced stronger information criteria results.

The ARMA(0,1)+GARCH(1,1) model was useful for investigating volatility clustering in the log-return series. It showed that periods of high variability occurred in the AI search interest data, especially around the sharp increases after 2022. However, because GARCH focuses on conditional variance rather than directly forecasting the original search-interest level, it was treated as a supporting volatility model rather than the final forecasting model.

Overall, SARIMA$(0,1,1)\times(0,1,1)_{12}$ was selected as the final model because it provided the best balance between statistical fit, interpretability, seasonal structure, and forecasting usefulness.

3.9 Forecasting Results

The final forecasts from the SARIMA$(0,1,1)\times(0,1,1)_{12}$ model indicate that global search interest in Artificial Intelligence is expected to remain at historically high levels from June 2026 to March 2027. The first two forecast months, June and July 2026, are predicted to remain around 82, which is still much higher than the historical average of the series. From August 2026 to January 2027, the forecasts remain close to 97–98, suggesting that public interest in AI is expected to remain near its recent peak.

The February and March 2027 forecasts exceed 100, with point forecasts of 114.60 and 112.00 respectively. Since the Google Trends index is capped at 100, these values should be interpreted carefully. They do not mean that the actual Google Trends index can exceed 100. Instead, they indicate that the statistical model expects AI search interest to remain extremely strong, close to the maximum possible Google Trends level.

The prediction intervals widen as the forecast horizon increases, showing that uncertainty grows further into the future. This is expected in time series forecasting because predictions become less certain as they move further away from the observed data. Despite this uncertainty, the forecast results consistently indicate that AI search interest is unlikely to return to its earlier low levels in the short term. Therefore, the forecasting results support the conclusion that public attention toward Artificial Intelligence is expected to remain strong through early 2027.

4. DISCUSSION

This study applied several time series modelling techniques to worldwide Google Trends search interest for the term “artificial intelligence”. The descriptive analysis showed that the series was strongly non-stationary, highly persistent, and dominated by a sharp increase in interest after late 2022. This change coincides with the public release of ChatGPT and the broader rise of generative AI tools. The results suggest that public attention toward AI has undergone a major structural shift rather than following a simple stable pattern over time.

The trend models showed that a simple linear trend was not sufficient to explain the data. The linear model underestimated the rapid increase in recent years and failed to capture the nonlinear nature of AI search interest. The quadratic and seasonal quadratic models performed better because they allowed for a curved long-term trend. However, regression-based trend models alone are limited because they do not fully model the autocorrelation structure of time series data.

The ARIMA model improved the analysis by modelling dependence between observations after differencing. The first-differenced series appeared more stationary, and the ARIMA(0,1,1) model provided a reasonable non-seasonal forecasting benchmark. However, the monthly structure of the data suggested that seasonal dependence should also be considered. This led to the development of SARIMA models.

The SARIMA$(0,1,1)\times(0,1,1)_{12}$ model was selected as the final forecasting model because it accounted for both non-seasonal and seasonal components. It produced strong information criteria results and reasonable forecast accuracy. The model forecasted that AI search interest would remain high over the next 10 months, with several forecasts close to or above the maximum Google Trends index value. This supports the view that AI interest is likely to remain strong in the short term.

The GARCH analysis provided an additional perspective by examining changes in volatility. The return series showed periods of higher variability, especially during the recent rapid growth phase. The ARMA(0,1)+GARCH(1,1) model captured this time-varying volatility and showed that uncertainty in AI search interest has not remained constant over time. However, GARCH modelling was more useful for understanding volatility than for producing the final forecasts of the original search-interest index.

There are several limitations to this analysis. First, Google Trends data are normalised, meaning that the values are relative to the highest search volume in the selected period rather than absolute search counts. Second, the series contains a major structural break after 2022, which makes forecasting more difficult because the future may not follow the same pattern as the earlier historical data. Third, the Google Trends index is bounded between 0 and 100, so forecasts above 100 must be interpreted as indicating interest near the maximum level rather than literal values. Finally, external factors such as new AI product releases, government regulation, media attention, or economic conditions may influence future search interest but are not explicitly included in the univariate time series models used in this study.

Despite these limitations, the analysis demonstrates that SARIMA modelling is suitable for short-term forecasting of monthly AI search interest. It also shows that combining descriptive analysis, stationarity testing, model comparison, residual diagnostics, and forecast interpretation provides a structured and reliable approach to analysing real-world time series data.

5. CONCLUSION

This report investigated worldwide Google Trends search interest for “artificial intelligence” from January 2004 to May 2026 and developed time series models to forecast interest for the next 10 months. The analysis showed that the series was non-stationary, highly autocorrelated, and strongly affected by rapid growth after late 2022.

Several model families were considered, including trend regression, ARIMA, SARIMA, and ARMA-GARCH models. The trend models captured the broad long-term increase but were limited in modelling autocorrelation. The ARIMA model provided a useful non-seasonal benchmark, while the GARCH model helped explain volatility clustering in the return series. However, the SARIMA(0,1,1)×(0,1,1)$_{12}$ model was selected as the best overall forecasting model because it captured both non-seasonal and seasonal patterns in the monthly data.

The final SARIMA forecasts suggest that global interest in Artificial Intelligence will remain very high from June 2026 to March 2027. Forecasts remain around 82 in the first two months and rise close to or above the upper range of the Google Trends index in later months. Since the index is bounded at 100, forecasts above 100 indicate that AI search interest is expected to remain near the maximum observed level rather than literally exceeding the scale.

Overall, the research question is answered by concluding that the SARIMA(0,1,1)×(0,1,1)$_{12}$ model is the most suitable model among those considered for forecasting worldwide AI search interest. The results indicate that public interest in Artificial Intelligence is likely to remain strong in the short-term future.

6. APPENDIX

APPENDIX A. RESIDUAL DIAGNOSTIC FRAMEWORK

The adequacy of each fitted model was evaluated using a standard residual diagnostic framework. Residual analysis is an essential step in time series modelling because an appropriate model should leave residuals that behave like white noise. White-noise residuals indicate that all systematic information contained in the original series has been successfully captured by the model.

The following diagnostic procedures were applied:

1. Residual Time Series Plot

Residuals were plotted against time to assess whether they fluctuated randomly around zero. The absence of systematic patterns, trends, or changing variance provides evidence that the model has adequately captured the underlying structure of the data.

2. Histogram of Residuals

Histograms were examined to assess the distribution of residuals. A bell-shaped distribution suggests approximate normality and supports the validity of statistical inference.

3. Normal Q-Q Plot

Quantile-Quantile plots were used to compare the empirical distribution of residuals with the theoretical normal distribution. Points lying close to the reference line indicate approximate normality.

4. Autocorrelation Function (ACF)

Residual autocorrelations were examined to determine whether significant serial dependence remained after model fitting. Residual autocorrelations within the 95% confidence limits indicate that the model has successfully removed the temporal dependence structure.

5. Ljung-Box Test

The Ljung-Box test was applied to formally assess whether residuals exhibited significant autocorrelation.

H_0: Residuals are independently distributed (white noise)

H_1: Residuals are autocorrelated

A p-value greater than 0.05 indicates insufficient evidence to reject the null hypothesis and suggests that the residuals behave as white noise.

6. Shapiro-Wilk Test

The Shapiro-Wilk test was used to evaluate the normality assumption.

H_0: Residuals follow a normal distribution

H_1: Residuals do not follow a normal distribution

A p-value greater than 0.05 indicates that normality cannot be rejected.

7. McLeod-Li Test

For volatility models, the McLeod-Li test was used to assess whether any remaining ARCH effects were present in the residuals. A non-significant result indicates that conditional heteroscedasticity has been adequately modelled.

Overall, a model was considered adequate when the residual series displayed no systematic patterns, the residual ACF contained no significant spikes, and the formal diagnostic tests supported the white-noise assumption.

APPENDIX B. SOFTWARE AND PACKAGE INFORMATION

The analysis was conducted using R version 4.5.0 within the RStudio integrated development environment.

The following packages were used throughout the analysis:

• TSA – Time series modelling and diagnostics

• forecast – Forecast generation and ARIMA modelling

• tseries – Stationarity testing

• fUnitRoots – Unit root testing

• lmtest – Statistical hypothesis testing

• fGarch – GARCH model estimation

• rugarch – Advanced GARCH modelling

• Hmisc – Data manipulation and utilities

• stats – Regression modelling and statistical procedures

• graphics – Data visualisation

• grDevices – Graphical utilities

The analysis was performed using RStudio, and all computations, model estimation procedures, graphical outputs, diagnostic checks, and forecasts were generated using R.

APPENDIX C. COMPLETE R CODE

````
--- title: "Forecasting Global Search Interest in Artificial
Intelligence" subtitle: "ARIMA, SARIMA, and GARCH Modelling of Global AI
Search Interest" author: |
  | Jayasri Ravichandran (S4144181)
  | Irine Biju Panicker (S4137661)
  | Aiswarya Sibi (S4133705)
  | Group 66 date: "June
2026" output:
pdf_document:     toc:
true
number_sections: true
html_document:     toc:
true     toc_float: true
toc_depth: 3     theme:
flatly     highlight:
tango
number_sections: true
df_print: kable
---

```{r global-options,
include=FALSE}
knitr::opts_chunk$set(   echo
= TRUE,
  message   = FALSE,
warning   = FALSE,
fig.width = 10,
fig.height = 6,
fig.align = "center"
)
```

---
````

```
**Research Question:** What are the most accurate 10-month forecasts for
global AI search interest (Google Trends) for the period June 2026 to March
2027?

**Data:** Google Trends monthly search interest index for
"artificial intelligence" worldwide, January 2004 – May 2026 (n =
269).
---



## Libraries and Helper Functions

```{r setup, message=FALSE, warning=FALSE, echo=TRUE}
# ---- 0. LIBRARIES & HELPER FUNCTIONS
-----------------------------------------
 rm(list =
ls())
 library(TSA)
library(fUnitRoots
)
library(forecast)
library(lmtest)
library(tseries)
library(fGarch)
library(rugarch)
library(Hmisc)

# --- sort.score() ---
sort.score <- function(x, score = c("bic", "aic"))
{   if (score == "aic") {     x[with(x,
order(AIC)), ]   } else if (score == "bic") {
x[with(x, order(BIC)), ]
  } else {
    warning('score = "x" only accepts valid arguments ("aic","bic")')
}
}

# --- safe.AIC() / safe.BIC() — handles CSS-fallback models
-------------------
get_ic <- function(model, type = "AIC") {
val <- tryCatch(
    if (type == "AIC") AIC(model) else BIC(model),
error = function(e) numeric(0)
  )
  if (length(val) == 0 || is.na(val)) return(NA_real_)
as.numeric(val)
}  safe.AIC <-
function(...) {   models
<- list(...)
  nms    <- sapply(match.call()[-1], deparse)
  vals   <- vapply(models, get_ic, numeric(1), type = "AIC")
result <- data.frame(AIC = vals, row.names = nms)
result[order(result$AIC, na.last = TRUE), , drop = FALSE]
}
safe.BIC <- function(...) {
models <- list(...)
```
```

```
  nms    <- sapply(match.call()[-1], deparse)
  vals   <- vapply(models, get_ic, numeric(1), type = "BIC")
result <- data.frame(BIC = vals, row.names = nms)
result[order(result$BIC, na.last = TRUE), , drop = FALSE]
}
# --- ljung.box.plot() — replaces FitAR::LBQPlot, no extra package needed
-----
ljung.box.plot <- function(res, lag.max = 30) {
lags  <- 1:lag.max
  pvals <- sapply(lags, function(i) Box.test(res, lag = i, type =
"Ljung-Box")$p.value)
  plot(lags, pvals, xlab = "Lag", ylab = "p-value", ylim = c(0, 1),
main = "Ljung-Box Q-test p-values",
       pch = 20, col = ifelse(pvals < 0.05, "red", "steelblue"))
abline(h = 0.05, col = "red", lty = 2)
  legend("topright", legend = c("p-value", "5% level"),
         col = c("steelblue", "red"), pch = c(20, NA), lty = c(NA, 2))
}

# --- residual.analysis()
-----------------------------------------------------
residual.analysis <- function(model, std = TRUE, start = 2, lag.max = 30,
class = c("ARIMA", "SARIMA", "GARCH",
"ARMA-GARCH", "garch", "fGARCH")[1])
{
library(TSA)
  if ((class == "ARIMA") | (class == "SARIMA")) {
    res.model <- if (std) rstandard(model) else residuals(model)
  } else if (class == "GARCH") {
    res.model <- model$residuals[start:model$n.used]
  } else if (class == "garch") {
    res.model <- model$residuals[start:model$n.used]
  } else if (class == "ARMA-GARCH") {
res.model <- model@fit$residuals   }
else if (class == "fGARCH") {
res.model <- model@residuals
  } else {
    stop("'class' must be one of: ARIMA, SARIMA, GARCH, garch, ARMA-GARCH,
fGARCH")   }   res.plain <- as.numeric(res.model)   # fix fractional lag
bug in ACF   layout(matrix(c(1,2,3,4,5,6), nrow = 3, ncol = 2, byrow =
TRUE))   par(mar = c(4, 4, 3, 1))
  plot(res.plain, type = "o", ylab = "Standardised Residuals",
main = "Time Series Plot of Standardised Residuals")   abline(h =
0)
  hist(res.plain, main = "Histogram of Standardised Residuals",
xlab = "Standardised Residuals")
  qqnorm(res.plain, main = "QQ Plot of Standardised
Residuals")   qqline(res.plain, col = 2)   if (class ==
"SARIMA") {
    seasonal_acf(res.plain, main = "ACF of Standardised Residuals")
} else {
    acf(res.plain, main = "ACF of Standardised Residuals")
  }
  print(shapiro.test(res.plain))
  ljung.box.plot(res.plain, lag.max = lag.max)
layout(1)
  par(mar = c(5, 4, 4, 2) + 0.1)
}
```

```
# --- seasonal_acf() / seasonal_pacf() --- helper
<- function(class = c("acf", "pacf"), ...) {
params <- match.call(expand.dots = TRUE)   params
<- as.list(params)[-1]
  params[["class"]] <- NULL   # remove 'class' so it is not passed
to acf()/pacf()   if (class == "acf") {
    acf_values <- do.call(acf, c(params, list(plot = FALSE)))
  } else {     acf_values <- do.call(pacf, c(params,
list(plot = FALSE)))
  }   acf_data <- data.frame(Lag =
as.numeric(acf_values$lag),                          ACF
= as.numeric(acf_values$acf))   seasonal_lags <-
acf_data$Lag %% 1 == 0   if (class == "acf") {
    do.call(acf, c(params, list(plot = TRUE)))
  } else {
    do.call(pacf, c(params, list(plot = TRUE)))
  }   for (i in which(seasonal_lags)) {
segments(x0 = acf_data$Lag[i], y0 = 0,
             x1 = acf_data$Lag[i], y1 = acf_data$ACF[i], col =
"red")   } }
seasonal_acf  <- function(...) helper(class = "acf",  ...)
seasonal_pacf <- function(...) helper(class = "pacf", ...)


#
============================================================================
= ```
```

1. Data Loading and Preparation

```
```{r data, echo=TRUE}
# 1.  DATA LOADING & PREPARATION
#
============================================================================
=

raw <- read.csv("multiTimeline_AI_Growth.csv", skip = 1, header =
TRUE) colnames(raw) <- c("Month", "AI_Interest") head(raw) tail(raw)
dim(raw)   # 269 observations

# Convert to ts object: monthly, starting January 2004 ai.ts
<- ts(raw$AI_Interest, start = c(2004, 1), frequency = 12)
ai.ts

#
============================================================================
= ```
```

2. Descriptive Analysis

```
```{r descriptive, echo=TRUE, fig.width=10, fig.height=6}
# 2.  DESCRIPTIVE ANALYSIS
#
============================================================================
=
```

```
# --- 2.1 Time series plot ---
plot(ai.ts, type = "o",
     ylab = "Search Interest Index (0–100)",
xlab = "Year",
     main = "Figure 1. Monthly Global Search Interest in 'Artificial
Intelligence'\n(Google Trends, Jan 2004 – May 2026)")

# Add reference line at ChatGPT launch (Nov 2022)
abline(v = 2022 + 10/12, col = "red", lty = 2)
legend("topleft", legend = c("AI Interest", "ChatGPT Launch (Nov
2022)"),        col = c("black", "red"), lty = c(1, 2), pch = c(1, NA))
# --- 2.2 Seasonal plot (month labels) --
- par(mar = c(5, 5, 4, 2)) plot(ai.ts,
type = "l",
     ylab = "Search Interest Index (0–100)",
xlab = "Year",
     main = "Figure 2. Monthly Search Interest with Seasonal Labels")
par(mar = c(5, 4, 4, 2))
points(y = ai.ts, x = time(ai.ts), pch =
as.vector(season(ai.ts)))
# --- 2.3 Summary statistics ---
summary(ai.ts)

# --- 2.4 Lag-1 scatter plot & correlation --
- y     <- ai.ts x     <- zlag(ai.ts) index
<- 2:length(x)
cat("Lag-1 correlation:", cor(y[index], x[index]),
"\n") plot(y[index], x[index],      xlab = "Search
Interest (t-1)",      ylab = "Search Interest (t)",
main = "Figure 3. Lag-1 Scatter Plot")

# --- 2.5 ACF & PACF of raw series ---
par(mfrow = c(1, 2), mar = c(5, 5, 4, 2))
seasonal_acf(ai.ts,  lag.max = 48,
             main = "Figure 4. ACF of AI Search
Interest",              xlab = "Lag")
seasonal_pacf(ai.ts, lag.max = 48,
              main = "Figure 5. PACF of AI Search Interest",
xlab = "Lag")
par(mfrow = c(1, 1), mar = c(5, 4, 4, 2)) #
Slow decay in ACF → clear non-stationarity


#
============================================================================
= ```

## 3. Trend Models (Regression)

```{r trend, echo=TRUE, fig.width=12, fig.height=8}
# 3.  TREND MODELS (REGRESSION)
#
============================================================================
=
```
```

```
t  <- time(ai.ts)
t2  <-  t^2  t3  <-
t^3

# --- Model 1: Linear trend --
- model.lin <- lm(ai.ts ~ t)
summary(model.lin) par(mar =
c(5, 5, 4, 2)) plot(ai.ts,
type = "o",      ylab =
"Search Interest",      xlab =
"Year",
     main = "Figure 6. Linear Trend
Model") abline(model.lin, col = "red")
par(mar = c(5, 4, 4, 2))
layout(matrix(1:4, nrow=2, ncol=2))
par(mar = c(4, 4, 3, 1)) res.tmp <-
rstudent(model.lin)
plot(res.tmp, x = as.vector(time(ai.ts)), type =
"l",      xlab = "Time", ylab = "Standardised
Residuals",      main = "Residuals - Linear Model")
hist(res.tmp, xlab = "Standardised Residuals",
main = "Histogram - Linear Model") qqnorm(res.tmp,
main = "QQ Plot - Linear Model") qqline(res.tmp,
col = 2)
acf(as.numeric(res.tmp), main = "ACF - Linear Model Residuals")
layout(1)
par(mar = c(5, 4, 4, 2) + 0.1)
shapiro.test(res.tmp)

# --- Model 2: Quadratic trend --
- model.quad <- lm(ai.ts ~ t +
t2) summary(model.quad) par(mar =
c(5, 5, 4, 2)) plot(ai.ts, type =
"o",
     ylim = range(c(fitted(model.quad),
as.vector(ai.ts))),      ylab = "Search Interest",
xlab = "Year",
     main = "Figure 7. Quadratic Trend Model")
lines(ts(fitted(model.quad), start = c(2004, 1), frequency = 12),
col = "red", lty = 2)
legend("topleft", lty = c(1, 2), col = c("black",
"red"),        legend = c("Observed", "Fitted"))
par(mar = c(5, 4, 4, 2)) layout(matrix(1:4, nrow=2,
ncol=2)) par(mar = c(4, 4, 3, 1)) res.tmp <-
rstudent(model.quad)
plot(res.tmp, x = as.vector(time(ai.ts)), type =
"l",      xlab = "Time", ylab = "Standardised
Residuals",      main = "Residuals - Quadratic
Model") hist(res.tmp, xlab = "Standardised
Residuals",      main = "Histogram - Quadratic
Model") qqnorm(res.tmp, main = "QQ Plot - Quadratic
Model") qqline(res.tmp, col = 2)
acf(as.numeric(res.tmp), main = "ACF - Quadratic Model Residuals")
layout(1)
par(mar = c(5, 4, 4, 2) + 0.1)
shapiro.test(res.tmp)
```

```
# --- Model 3: Cubic trend ---
model.cub <- lm(ai.ts ~ t + t2 +
t3) summary(model.cub) par(mar =
c(5, 5, 4, 2)) plot(ai.ts, type =
"o",
     ylim = range(c(fitted(model.cub),
as.vector(ai.ts))),      ylab = "Search Interest",
xlab = "Year",
     main = "Figure 8. Cubic Trend Model\n(Note: cubic fit identical to
quadratic — cubic coefficient not estimable)")
lines(ts(fitted(model.cub), start = c(2004, 1), frequency = 12),
col = "red", lty = 2)
legend("topleft", lty = c(1, 2), col = c("black",
"red"),        legend = c("Observed", "Fitted"))
par(mar = c(5, 4, 4, 2)) layout(matrix(1:4, nrow=2,
ncol=2)) par(mar = c(4, 4, 3, 1)) res.tmp <-
rstudent(model.cub)
plot(res.tmp, x = as.vector(time(ai.ts)), type =
"l",      xlab = "Time", ylab = "Standardised
Residuals",      main = "Residuals - Cubic Model")
hist(res.tmp, xlab = "Standardised Residuals",
main = "Histogram - Cubic Model") qqnorm(res.tmp,
main = "QQ Plot - Cubic Model") qqline(res.tmp, col
= 2)
acf(as.numeric(res.tmp), main = "ACF - Cubic Model Residuals")
layout(1)
par(mar = c(5, 4, 4, 2) + 0.1)
shapiro.test(res.tmp)

# --- Model 4: Seasonal means model --
- month. <- season(ai.ts) model.seas
<- lm(ai.ts ~ month. - 1)
summary(model.seas)
layout(matrix(1:4, nrow=2,
ncol=2)) par(mar = c(4, 4, 3, 1))
res.tmp <- rstudent(model.seas)
plot(res.tmp, x = as.vector(time(ai.ts)), type =
"l",      xlab = "Time", ylab = "Standardised
Residuals",      main = "Residuals - Seasonal
Model") hist(res.tmp, xlab = "Standardised
Residuals",      main = "Histogram - Seasonal
Model") qqnorm(res.tmp, main = "QQ Plot - Seasonal
Model") qqline(res.tmp, col = 2)
acf(as.numeric(res.tmp), main = "ACF - Seasonal Model Residuals")
layout(1)
par(mar = c(5, 4, 4, 2) + 0.1)
shapiro.test(res.tmp)

# --- Model 5: Seasonal + quadratic ---
model.seas.quad <- lm(ai.ts ~ month. + t + t2 -
1) summary(model.seas.quad) par(mar = c(5, 5, 4,
2)) plot(ai.ts, type = "o",
     ylim = range(c(fitted(model.seas.quad),
as.vector(ai.ts))),      ylab = "Search Interest",      xlab =
"Year",
     main = "Figure 9. Seasonal + Quadratic Trend Model")
lines(ts(fitted(model.seas.quad), start = c(2004, 1), frequency = 12),
col = "red", lty = 2)
```

```
legend("topleft", lty = c(1, 2), col = c("black",
"red"),        legend = c("Observed", "Fitted"))
par(mar = c(5, 4, 4, 2)) layout(matrix(1:4, nrow=2,
ncol=2)) par(mar = c(4, 4, 3, 1)) res.tmp <-
rstudent(model.seas.quad)
plot(res.tmp, x = as.vector(time(ai.ts)), type = "l",
xlab = "Time", ylab = "Standardised Residuals",      main =
"Residuals - Seasonal + Quadratic Model") hist(res.tmp, xlab
= "Standardised Residuals",      main = "Histogram -
Seasonal + Quadratic Model") qqnorm(res.tmp, main = "QQ Plot
- Seasonal + Quadratic Model") qqline(res.tmp, col = 2)
acf(as.numeric(res.tmp), main = "ACF - Seasonal + Quadratic Model
Residuals") layout(1)
par(mar = c(5, 4, 4, 2) + 0.1)
shapiro.test(res.tmp)

# --- Model 6: Harmonic --- har.
<- harmonic(ai.ts, 1) data.har   <-
data.frame(ai.ts, har.)
model.har  <- lm(ai.ts ~ cos.2.pi.t. + sin.2.pi.t., data = data.har)
summary(model.har)
layout(matrix(1:4, nrow=2,
ncol=2)) par(mar = c(4, 4, 3, 1))
res.tmp <- rstudent(model.har)
plot(res.tmp, x = as.vector(time(ai.ts)), type =
"l",      xlab = "Time", ylab = "Standardised
Residuals",      main = "Residuals - Harmonic
Model") hist(res.tmp, xlab = "Standardised
Residuals",      main = "Histogram - Harmonic
Model") qqnorm(res.tmp, main = "QQ Plot - Harmonic
Model") qqline(res.tmp, col = 2)
acf(as.numeric(res.tmp), main = "ACF - Harmonic Model Residuals")
layout(1)
par(mar = c(5, 4, 4, 2) + 0.1)
shapiro.test(res.tmp)

# --- AIC / BIC comparison for trend models ---
sort.score(AIC(model.lin, model.quad, model.cub,
               model.seas, model.seas.quad, model.har), score =
"aic") sort.score(AIC(model.lin, model.quad, model.cub,
model.seas, model.seas.quad, model.har,                k =
log(length(ai.ts))), score = "aic")

# --- Forecasts from best trend model (seasonal + quadratic) -
-- h              <- 10   # 10 months ahead last.t          <-
t[length(t)] dataFreq       <- frequency(ai.ts) new.t
<- seq(last.t + 1/dataFreq,
                      last.t + h/dataFreq, by = 1/dataFreq)
ahead.seas.quad <- data.frame(
  month.  =  c("June",  "July",  "August",  "September",  "October",
"November",  "December",  "January",  "February",  "March"),    t   =
new.t,   t2 = new.t^2
)
frc.seas.quad <- predict(model.seas.quad,
newdata = ahead.seas.quad,
interval = "prediction")  par(mar = c(5, 5, 4,
2)) plot(ai.ts,
```

```
     xlim = c(2004, last.t + h/dataFreq + 0.1),
ylim = c(0, 130),
     ylab = "Search Interest Index",
xlab = "Year",
     main = "Figure 10. 10-Month Forecasts – Seasonal + Quadratic Trend
Model")
lines(ts(fitted(model.seas.quad), start = c(2004,1), frequency = 12),
col = "green")
lines(ts(as.vector(frc.seas.quad[, 1]), start = c(2026, 6), frequency = 12),
col = "red")
lines(ts(as.vector(frc.seas.quad[, 2]), start = c(2026, 6), frequency = 12),
col = "blue")
lines(ts(as.vector(frc.seas.quad[, 3]), start = c(2026, 6), frequency =
12),       col = "blue") legend("topleft", lty = 1,
       col = c("black", "green", "red", "blue"),
legend = c("Data", "Fitted", "Forecasts", "95% PI"))
par(mar = c(5, 4, 4, 2))


#
=============================================================================
= ```

## 4. ARIMA Modelling

```{r arima, echo=TRUE, fig.width=12, fig.height=10}
# 4.  ARIMA MODELLING
#
=============================================================================
=
# --- 4.1 Normality & Box-Cox transformation -
-- par(mar = c(5, 5, 4, 2)) qqnorm(ai.ts,
       main = "Figure 11. QQ Plot – Raw
Series",        xlab = "Theoretical
Quantiles",        ylab = "Sample Quantiles")
qqline(ai.ts, col = 2) par(mar = c(5, 4, 4,
2)) shapiro.test(ai.ts)

# BoxCox.ar from TSA — professor's method
BC    <- BoxCox.ar(ai.ts)
title(main = "Figure 11a. Box-Cox Log-Likelihood Profile")
BC$ci
lambda <- BC$lambda[which(max(BC$loglike) == BC$loglike)]
lambda
BC.ai <- (ai.ts^lambda - 1) / lambda
 par(mar = c(5, 5, 4,
2)) plot(BC.ai, type =
"o",
     ylim = c(floor(min(BC.ai) * 10) / 10 - 0.05,
ceiling(max(BC.ai) * 10) / 10 + 0.05),      ylab =
"BC Transformed Index",      xlab = "Year",
     main = "Figure 12. Box-Cox Transformed Series")
par(mar = c(5, 4, 4, 2))

par(mar = c(5, 5, 4, 2))
qqnorm(BC.ai,
```
```

```
       main = "Figure 13. QQ Plot - BC Transformed
Series",        xlab = "Theoretical Quantiles",
ylab = "Sample Quantiles") qqline(BC.ai, col = 2) par(mar
= c(5, 4, 4, 2)) shapiro.test(BC.ai)

# --- 4.2 Stationarity tests on raw series --
- adf.test(ai.ts) pp.test(ai.ts)

# --- 4.3 First difference --- diff1.ai
<- diff(ai.ts, differences = 1) par(mar
= c(5, 5, 4, 2)) plot(diff1.ai, type =
"o",
     ylim = c(min(diff1.ai) - 2, max(diff1.ai) +
2),      ylab = "First Difference",      xlab =
"Year",
     main = "Figure 14. First Difference of AI Search Interest")
# Note: large spikes (~+30 and -25) visible in 2025-2026 reflect the ChatGPT
# structural break — acknowledge these outliers in Section 3.3 report text.
par(mar = c(5, 4, 4, 2))

adf.test(diff1.ai)
pp.test(diff1.ai)
kpss.test(diff1.ai
)
# NOTE for report text: report KPSS result here — "The KPSS test further
confirmed
# stationarity of the differenced series (p > 0.10)." Also note that ARIMA
models
# are fitted to the original series (not Box-Cox transformed) as the near-
log # transform did not substantially alter the autocorrelation structure.

par(mfrow = c(1, 2), mar = c(5, 5, 4, 2))
acf(diff1.ai,  lag.max = 48,
    main = "Figure 15. ACF - First
Difference",     xlab = "Lag") pacf(diff1.ai,
lag.max = 48,
     main = "Figure 16. PACF - First Difference",
xlab = "Lag")
par(mfrow = c(1, 1), mar = c(5, 4, 4, 2))

# --- 4.4 Second difference (if needed) --
- diff2.ai <- diff(ai.ts, differences =
2) par(mar = c(5, 5, 4, 2))
plot(diff2.ai, type = "o",
     ylim = c(min(diff2.ai) - 2, max(diff2.ai) +
2),      ylab = "Second Difference",      xlab =
"Year",
     main = "Figure 17. Second Difference of AI Search Interest")
par(mar = c(5, 4, 4, 2))

adf.test(diff2.ai)
pp.test(diff2.ai)
kpss.test(diff2.ai
)

par(mfrow = c(1, 2), mar = c(5, 5, 4, 2))
acf(diff2.ai,  lag.max = 48,
```

```
    main = "Figure 18. ACF - Second Difference",
xlab = "Lag")
pacf(diff2.ai, lag.max = 48,
     main = "Figure 19. PACF - Second Difference",
xlab = "Lag")
par(mfrow = c(1, 1), mar = c(5, 4, 4, 2))

# d = 1 (use diff1.ai for model identification)

# --- 4.5 EACF ---
eacf(diff1.ai, ar.max = 6, ma.max = 6)

# --- 4.6 BIC subset selection ---
res.arma <- armasubsets(y = diff1.ai, nar = 6, nma = 6,
                         y.name = "p", ar.method = "ols")
plot(res.arma)

# Candidate models (update based on your ACF/PACF/EACF output):
# {ARIMA(0,1,1), ARIMA(1,1,1), ARIMA(2,1,1),
#  ARIMA(0,1,2), ARIMA(1,1,2), ARIMA(2,1,2)}

# --- 4.7 Fit candidate ARIMA models ---

# ARIMA(0,1,1)
m.011.css <- Arima(ai.ts, order = c(0,1,1), method = "CSS")
m.011.ml  <- tryCatch(
  Arima(ai.ts, order = c(0,1,1), method = "ML"),
error = function(e) tryCatch(
    Arima(ai.ts, order = c(0,1,1), method = "CSS-ML"),
    error = function(e) Arima(ai.ts, order = c(0,1,1), method = "CSS")
) )
coeftest(m.011.ml)
residual.analysis(m.011.ml, lag.max = 30)

# ARIMA(1,1,1)
m.111.css <- Arima(ai.ts, order = c(1,1,1), method = "CSS")
m.111.ml  <- tryCatch(
  Arima(ai.ts, order = c(1,1,1), method = "ML"),
error = function(e) tryCatch(
    Arima(ai.ts, order = c(1,1,1), method = "CSS-ML"),
    error = function(e) Arima(ai.ts, order = c(1,1,1), method = "CSS")
  ) )
coeftest(m.111.ml)
residual.analysis(m.111.ml, lag.max = 30)

# ARIMA(2,1,1)
m.211.css <- Arima(ai.ts, order = c(2,1,1), method = "CSS")
m.211.ml  <- tryCatch(
  Arima(ai.ts, order = c(2,1,1), method = "ML"),
error = function(e) tryCatch(
    Arima(ai.ts, order = c(2,1,1), method = "CSS-ML"),
    error = function(e) Arima(ai.ts, order = c(2,1,1), method = "CSS")
  ) )
coeftest(m.211.ml)
residual.analysis(m.211.ml, lag.max = 30)
```

```
# ARIMA(0,1,2)
m.012.css <- Arima(ai.ts, order = c(0,1,2), method = "CSS")
m.012.ml  <- tryCatch(
  Arima(ai.ts, order = c(0,1,2), method = "ML"),
  error = function(e) tryCatch(
    Arima(ai.ts, order = c(0,1,2), method = "CSS-ML"),
    error = function(e) Arima(ai.ts, order = c(0,1,2), method = "CSS")
) )
coeftest(m.012.ml)
residual.analysis(m.012.ml, lag.max = 30)

# ARIMA(1,1,2)
m.112.css <- Arima(ai.ts, order = c(1,1,2), method = "CSS")
m.112.ml  <- tryCatch(
  Arima(ai.ts, order = c(1,1,2), method = "ML"),
error = function(e) tryCatch(
    Arima(ai.ts, order = c(1,1,2), method = "CSS-ML"),
    error = function(e) Arima(ai.ts, order = c(1,1,2), method = "CSS")
) )
coeftest(m.112.ml)
residual.analysis(m.112.ml, lag.max = 30)

# ARIMA(2,1,2)
m.212.css <- Arima(ai.ts, order = c(2,1,2), method = "CSS")
m.212.ml  <- tryCatch(
  Arima(ai.ts, order = c(2,1,2), method = "ML"),
error = function(e) tryCatch(
    Arima(ai.ts, order = c(2,1,2), method = "CSS-ML"),
    error = function(e) Arima(ai.ts, order = c(2,1,2), method = "CSS")
) )
coeftest(m.212.ml)
residual.analysis(m.212.ml, lag.max = 30)

# --- 4.8 AIC / BIC comparison ---
safe.AIC(m.011.ml, m.111.ml, m.211.ml, m.012.ml, m.112.ml,
m.212.ml) safe.BIC(m.011.ml, m.111.ml, m.211.ml, m.012.ml,
m.112.ml, m.212.ml)
# --- 4.9 Accuracy table (CSS fits) ---
Sm.011 <- accuracy(m.011.css)[1:7]
Sm.111 <- accuracy(m.111.css)[1:7]
Sm.211 <- accuracy(m.211.css)[1:7]
Sm.012 <- accuracy(m.012.css)[1:7]
Sm.112 <- accuracy(m.112.css)[1:7]
Sm.212 <- accuracy(m.212.css)[1:7]
df.arima <- data.frame(rbind(Sm.011, Sm.111, Sm.211,
Sm.012, Sm.112, Sm.212))
colnames(df.arima)  <- c("ME","RMSE","MAE","MPE","MAPE","MASE","ACF1")
rownames(df.arima)  <- c("ARIMA(0,1,1)","ARIMA(1,1,1)","ARIMA(2,1,1)",
"ARIMA(0,1,2)","ARIMA(1,1,2)","ARIMA(2,1,2)") round(df.arima, digits =
3)

# --- 4.10 Overparameterisation check (replace with your best model) -
-- # Example: if ARIMA(0,1,1) is best, check ARIMA(1,1,1) and
ARIMA(0,1,2) coeftest(m.111.ml)   # AR(1) significant?
coeftest(m.012.ml)   # MA(2) significant?
```

```
# --- 4.11 ARIMA Forecasts (10 months ahead) --- #
Replace m.011.ml with your selected best ARIMA model
frc.arima <- forecast(m.011.ml, h = 10) par(mar =
c(5, 5, 5, 2)) plot(frc.arima,
     main = "Figure 20. 10-Month Forecasts – ARIMA Model",
     ylab = "Search Interest
Index",      xlab = "Year") par(mar
= c(5, 4, 4, 2)) frc.arima


#
============================================================================
= ```
```

5. SARIMA Modelling

```
```{r sarima, echo=TRUE, fig.width=12, fig.height=10}
# 5.  SARIMA MODELLING
#
============================================================================
=
# --- 5.1 Seasonal ACF/PACF of raw series ---
par(mfrow = c(1, 2), mar = c(5, 5, 4, 2))
seasonal_acf(ai.ts,  lag.max = 60,
             main = "Figure 21. Seasonal ACF – AI Search
Interest",              xlab = "Lag") seasonal_pacf(ai.ts,
lag.max = 60,
              main = "Figure 22. Seasonal PACF – AI Search Interest",
xlab = "Lag")
par(mfrow = c(1, 1), mar = c(5, 4, 4, 2))

# --- 5.2 Step 1: Apply seasonal difference D=1, s=12 ---
m.s1 <- Arima(ai.ts, order = c(0,0,0),
              seasonal = list(order = c(0,1,0), period =
12)) res.s1 <- residuals(m.s1) par(mar = c(5, 5, 4, 2))
plot(res.s1, type = "o",      ylab = "Residuals",      xlab
= "Year",
     main = "Figure 23. Residuals after Seasonal Difference")
par(mfrow = c(1, 2), mar = c(5, 5, 4, 2))
seasonal_acf(res.s1,  lag.max = 48, main = "ACF after D=1", xlab = "Lag")
seasonal_pacf(res.s1, lag.max = 48, main = "PACF after D=1", xlab = "Lag")
par(mfrow = c(1, 1), mar = c(5, 4, 4, 2))

# --- 5.3 Step 2: Add seasonal MA(1) ---
m.s2 <- Arima(ai.ts, order = c(0,0,0),
              seasonal = list(order = c(0,1,1), period =
12)) res.s2 <- residuals(m.s2) par(mar = c(5, 5, 4, 2))
plot(res.s2, type = "o",      ylab = "Residuals",      xlab
= "Year",
     main = "Figure 24. Residuals after SARIMA(0,1,1)_12")
par(mfrow = c(1, 2), mar = c(5, 5, 4, 2))
seasonal_acf(res.s2,  lag.max = 48, main = "ACF – SARIMA(0,1,1)_12
residuals", xlab = "Lag")
seasonal_pacf(res.s2, lag.max = 48, main = "PACF – SARIMA(0,1,1)_12
residuals", xlab = "Lag")
par(mfrow = c(1, 1), mar = c(5, 4, 4, 2))
```
```

```
# --- 5.4 Step 3: Add ordinary difference d=1 ---
m.s3 <- Arima(ai.ts, order = c(0,1,0),
              seasonal = list(order = c(0,1,1), period = 12))
res.s3 <- residuals(m.s3)
par(mar = c(5, 5, 4, 2))
plot(res.s3, type = "o",
ylab = "Residuals",
xlab = "Year",
     main = "Figure 25. Residuals after d=1, D=1")
par(mfrow = c(1, 2), mar = c(5, 5, 4, 2))
seasonal_acf(res.s3,  lag.max = 48, main = "ACF – d=1, D=1 residuals", xlab
=
"Lag")
seasonal_pacf(res.s3, lag.max = 48, main = "PACF – d=1, D=1 residuals", xlab
= "Lag")
par(mfrow = c(1, 1), mar = c(5, 4, 4,
2)) adf.test(res.s3) pp.test(res.s3)

# --- 5.5 Identify ordinary ARMA orders from residuals -
-- eacf(res.s3, ar.max = 5, ma.max = 5) layout(1)
bic.sarima <- armasubsets(y = res.s3, nar = 5, nma = 5,
                           y.name = "p", ar.method = "ols")
plot(bic.sarima)

# Candidate SARIMA models:
# {SARIMA(0,1,1)x(0,1,1)_12,
#  SARIMA(1,1,1)x(0,1,1)_12,
#  SARIMA(0,1,2)x(0,1,1)_12,
#  SARIMA(1,1,2)x(0,1,1)_12}

# --- 5.6 Fit candidate SARIMA models ---

#      SARIMA(0,1,1)x(0,1,1)_12
sm.011_011.ml     <-  tryCatch(
Arima(ai.ts, order = c(0,1,1),
        seasonal = list(order = c(0,1,1), period = 12), method =
"ML"),   error = function(e) tryCatch(     Arima(ai.ts, order =
c(0,1,1),
          seasonal = list(order = c(0,1,1), period = 12), method = "CSS-
ML"),     error = function(e) Arima(ai.ts, order = c(0,1,1),
          seasonal = list(order = c(0,1,1), period = 12), method =
"CSS")   ) )
sm.011_011.css <- Arima(ai.ts, order = c(0,1,1),
                         seasonal = list(order = c(0,1,1), period =
12),                          method = "CSS") coeftest(sm.011_011.ml)
residual.analysis(sm.011_011.ml, lag.max = 30, class =
"SARIMA")
#      SARIMA(1,1,1)x(0,1,1)_12
sm.111_011.ml     <-  tryCatch(
Arima(ai.ts, order = c(1,1,1),
        seasonal = list(order = c(0,1,1), period = 12), method =
"ML"),   error = function(e) tryCatch(     Arima(ai.ts, order =
c(1,1,1),
          seasonal = list(order = c(0,1,1), period = 12), method = "CSS-
ML"),     error = function(e) Arima(ai.ts, order = c(1,1,1),
          seasonal = list(order = c(0,1,1), period = 12), method =
"CSS")   ) )
sm.111_011.css <- Arima(ai.ts, order = c(1,1,1),
```

```
                         seasonal = list(order = c(0,1,1), period =
12),                          method = "CSS") coeftest(sm.111_011.ml)
residual.analysis(sm.111_011.ml, lag.max = 30, class =
"SARIMA")
#       SARIMA(0,1,2)x(0,1,1)_12
sm.012_011.ml     <-   tryCatch(
Arima(ai.ts, order = c(0,1,2),
        seasonal = list(order = c(0,1,1), period = 12), method =
"ML"),   error = function(e) tryCatch(     Arima(ai.ts, order =
c(0,1,2),
          seasonal = list(order = c(0,1,1), period = 12), method = "CSS-
ML"),     error = function(e) Arima(ai.ts, order = c(0,1,2),
          seasonal = list(order = c(0,1,1), period = 12), method =
"CSS")    ) )
sm.012_011.css <- Arima(ai.ts, order = c(0,1,2),
                         seasonal = list(order = c(0,1,1), period =
12),                          method = "CSS") coeftest(sm.012_011.ml)
residual.analysis(sm.012_011.ml, lag.max = 30, class =
"SARIMA")
#       SARIMA(1,1,2)x(0,1,1)_12
sm.112_011.ml     <-   tryCatch(
Arima(ai.ts, order = c(1,1,2),
        seasonal = list(order = c(0,1,1), period = 12), method =
"ML"),   error = function(e) tryCatch(     Arima(ai.ts, order =
c(1,1,2),
          seasonal = list(order = c(0,1,1), period = 12), method = "CSS-
ML"),     error = function(e) Arima(ai.ts, order = c(1,1,2),
          seasonal = list(order = c(0,1,1), period = 12), method =
"CSS")    ) )
sm.112_011.css <- Arima(ai.ts, order = c(1,1,2),
                         seasonal = list(order = c(0,1,1), period =
12),                          method = "CSS") coeftest(sm.112_011.ml)
residual.analysis(sm.112_011.ml, lag.max = 30, class =
"SARIMA")
# --- 5.7 AIC / BIC comparison ---
safe.AIC(sm.011_011.ml, sm.111_011.ml, sm.012_011.ml,
sm.112_011.ml) safe.BIC(sm.011_011.ml, sm.111_011.ml,
sm.012_011.ml, sm.112_011.ml)
# --- 5.8 Accuracy table ---
Ssm.011_011 <- accuracy(sm.011_011.css)[1:7]
Ssm.111_011 <- accuracy(sm.111_011.css)[1:7]
Ssm.012_011 <- accuracy(sm.012_011.css)[1:7] Ssm.112_011 <-
accuracy(sm.112_011.css)[1:7] df.sarima <-
data.frame(rbind(Ssm.011_011, Ssm.111_011,
Ssm.012_011, Ssm.112_011)) colnames(df.sarima) <-
c("ME","RMSE","MAE","MPE","MAPE","MASE","ACF1") rownames(df.sarima)
<- c("SARIMA(0,1,1)x(0,1,1)_12",
"SARIMA(1,1,1)x(0,1,1)_12",
                          "SARIMA(0,1,2)x(0,1,1)_12",
"SARIMA(1,1,2)x(0,1,1)_12") round(df.sarima, digits = 3)

# --- 5.9 Overparameterisation check --- #
Example: if SARIMA(0,1,1)x(0,1,1)_12 is best:
sm.011_011.over1 <- tryCatch(
Arima(ai.ts, order = c(1,1,1),
        seasonal = list(order = c(0,1,1), period = 12), method =
"ML"),   error = function(e) tryCatch(     Arima(ai.ts, order =
c(1,1,1),
```

```
          seasonal = list(order = c(0,1,1), period = 12), method = "CSS-
ML"),     error = function(e) Arima(ai.ts, order = c(1,1,1),
          seasonal = list(order = c(0,1,1), period = 12), method = "CSS")
) ) coeftest(sm.011_011.over1)   # AR(1) significant?

sm.011_011.over2 <- tryCatch(
Arima(ai.ts, order = c(0,1,2),
        seasonal = list(order = c(0,1,1), period = 12), method =
"ML"),   error = function(e) tryCatch(     Arima(ai.ts, order =
c(0,1,2),
          seasonal = list(order = c(0,1,1), period = 12), method = "CSS-
ML"),     error = function(e) Arima(ai.ts, order = c(0,1,2),
          seasonal = list(order = c(0,1,1), period = 12), method = "CSS")
) ) coeftest(sm.011_011.over2)   # MA(2) significant?

# --- 5.10 SARIMA Forecasts ---
# Replace sm.011_011.ml with your selected best SARIMA
model frc.sarima <- forecast(sm.011_011.ml, h = 10)
par(mar = c(5, 5, 5, 2)) plot(frc.sarima,
     main = "Figure 26. 10-Month Forecasts - SARIMA
Model",      ylab = "Search Interest Index",      xlab =
"Year") par(mar = c(5, 4, 4, 2)) frc.sarima



#
============================================================================
= ```

## 6. GARCH / ARMA+GARCH Modelling

```{r garch, echo=TRUE, fig.width=12, fig.height=10}
# 6.  GARCH / ARMA+GARCH MODELLING
#
============================================================================
=
# --- 6.1 Compute returns (first difference of log — use for GARCH) ---
# Because ai.ts has zeros, shift by +1 before log
ai.shifted  <- ai.ts + 1
ai.returns  <- diff(log(ai.shifted))
 par(mar = c(5, 5, 4, 2))
plot(ai.returns, type =
"o",      ylab = "Log
Returns",      xlab =
"Year",
     main = "Figure 27. Returns of AI Search Interest (diff of log)")
par(mar = c(5, 4, 4, 2))

# --- 6.2 McLeod-Li test for ARCH effects ---
par(mar = c(5, 5, 4, 2))
McLeod.Li.test(y = ai.returns,
               main = "Figure 28. McLeod-Li Test - AI Returns")
par(mar = c(5, 4, 4, 2))
# Significant lags → ARCH effects present → GARCH
needed
```
```

```
#
  --- 6.3 ACF/PACF of returns ---
par(mfrow = c(1, 2), mar = c(5, 5, 4, 2))
acf(ai.returns,  main = "Figure 29. ACF of Returns", xlab = "Lag")
pacf(ai.returns, main = "Figure 30. PACF of Returns", xlab = "Lag")
par(mfrow = c(1, 1), mar = c(5, 4, 4, 2))

adf.test(ai.returns)   # Should be stationary

# --- 6.4 ARMA specification for mean equation ---
eacf(ai.returns, ar.max = 5, ma.max = 5)
res.arma2 <- armasubsets(y = ai.returns, nar = 5, nma = 5,
y.name = "p", ar.method = "ols") plot(res.arma2)

# --- 6.5 Fit ARMA models for mean equation --- # (Keep this
simple — the GARCH part handles variance) ma.011.css <-
Arima(ai.returns, order = c(0,0,1), method = "CSS") ma.011.ml  <-
tryCatch(
  Arima(ai.returns, order = c(0,0,1), method = "ML"),
error = function(e) tryCatch(
    Arima(ai.returns, order = c(0,0,1), method = "CSS-ML"),
    error = function(e) Arima(ai.returns, order = c(0,0,1), method = "CSS")
) )
coeftest(ma.011.ml)
residual.analysis(ma.011.ml, lag.max = 30)

ma.111.css <- Arima(ai.returns, order = c(1,0,1), method = "CSS")
ma.111.ml  <- tryCatch(
  Arima(ai.returns, order = c(1,0,1), method = "ML"),
error = function(e) tryCatch(
    Arima(ai.returns, order = c(1,0,1), method = "CSS-ML"),
    error = function(e) Arima(ai.returns, order = c(1,0,1), method = "CSS")
) )
coeftest(ma.111.ml)
residual.analysis(ma.111.ml, lag.max = 30)
 safe.AIC(ma.011.ml,
ma.111.ml)
safe.BIC(ma.011.ml,
ma.111.ml)

# --- 6.6 Check ARCH effects in best ARMA residuals -
-- best.arma.res    <- ma.011.ml$residuals abs.res
<- abs(best.arma.res) sq.res           <-
best.arma.res^2

par(mfrow = c(1, 2), mar = c(5, 5, 4, 2))
acf(abs.res,  ci.type = "ma", main = "Figure 31. ACF – |Residuals|", xlab =
"Lag")
pacf(abs.res, main = "Figure 32. PACF – |Residuals|", xlab =
"Lag") par(mfrow = c(1, 1), mar = c(5, 4, 4, 2)) eacf(abs.res,
ar.max = 5, ma.max = 5)

par(mfrow = c(1, 2), mar = c(5, 5, 4, 2))
acf(sq.res,  ci.type = "ma", main = "Figure 33. ACF – Residuals²", xlab =
"Lag")
```

```
#
pacf(sq.res, main = "Figure 34. PACF – Residuals²", xlab =
"Lag") par(mfrow = c(1, 1), mar = c(5, 4, 4, 2)) eacf(sq.res,
ar.max = 5, ma.max = 5)

# --- 6.7 Fit GARCH models ---
  Candidates (update from EACF): {GARCH(0,1), GARCH(1,1), GARCH(1,2)}

g.01 <- garch(best.arma.res, order = c(0,1), trace = FALSE)
summary(g.01)
residual.analysis(g.01, class = "garch")

g.11 <- garch(best.arma.res, order = c(1,1), trace = FALSE)
summary(g.11)
residual.analysis(g.11, class = "garch")

g.12 <- garch(best.arma.res, order = c(1,2), trace = FALSE)
summary(g.12)
residual.analysis(g.12, start = 3, class = "garch")

sc.AIC.garch <- AIC(g.01, g.11, g.12)
sc.BIC.garch <- AIC(g.01, g.11, g.12, k =
log(length(best.arma.res))) sort.score(sc.AIC.garch, score = "aic")
sort.score(sc.BIC.garch, score = "aic")

# --- 6.8 Fit combined ARMA+GARCH using rugarch ---
# NOTE: garchOrder = c(q, p) in rugarch (reversed from garch())
# Update armaOrder and garchOrder to match your best models above

spec.ag <- ugarchspec(
  variance.model = list(model = "sGARCH",
                      # GARCH(0,1) → c(1,0); GARCH(1,1) → c(1,1)
garchOrder = c(1, 1)),
  mean.model     = list(armaOrder = c(0, 1),     # ARMA(0,1) =
MA(1)                         include.mean = FALSE),
distribution.model = "norm"
)
fit.ag <- ugarchfit(spec = spec.ag, data = ai.returns)
fit.ag
residual.analysis(fit.ag, class = "ARMA-GARCH")

# --- 6.9 AIC/BIC for ARMA+GARCH ---
AICs.ag <- infocriteria(fit.ag)[1]
BICs.ag <- infocriteria(fit.ag)[2]
cat("ARMA+GARCH AIC:", AICs.ag, "  BIC:", BICs.ag, "\n")

# Plot conditional variance
plot(fit.ag, which = 2)

# --- 6.10 GARCH Forecasts (10 steps ahead) ---
forc.ag <- ugarchforecast(fit.ag, data = ai.returns, n.ahead = 10)
plot(forc.ag, which = 1)   # Series forecasts
plot(forc.ag, which = 3)   # Conditional variance forecasts
forc.ag
```

```
#

#
================================================================================
== ```

## 7. Final Model Comparison

```{r comparison, echo=TRUE}
# 7.  MODEL COMPARISON SUMMARY
#
================================================================================
==  Compare best model from each family

cat("\n===== FINAL MODEL COMPARISON =====\n")

# Trend model accuracy (CSS-equivalent via lm)
cat("Best Trend Model: Seasonal + Quadratic\n")
cat("AIC:", AIC(model.seas.quad), "\n")
cat("BIC:", BIC(model.seas.quad), "\n")
 #
ARIMA
cat("\nBest ARIMA model:\n")
# e.g. sort.score output will tell you — replace m.011.ml below
cat("AIC:", AIC(m.011.ml), "  BIC:", BIC(m.011.ml), "\n")
accuracy(m.011.css)
 #
SARIMA
cat("\nBest SARIMA model:\n")
cat("AIC:", AIC(sm.011_011.ml), "  BIC:", BIC(sm.011_011.ml), "\n")
accuracy(sm.011_011.css)

# ARMA+GARCH
cat("\nARMA+GARCH:\n")
cat("AIC:", infocriteria(fit.ag)[1],      "
BIC:", infocriteria(fit.ag)[2], "\n")

 ```
```

APPENDIX D. AI USAGE DECLARATION

AI Tool Used OpenAI ChatGPT (GPT-4o)
How AI Was Used
Artificial Intelligence tools were used only to assist with understanding the assessment requirements and improving the formatting and structure of the report.
Specifically, AI was used for:
• Understanding the assignment instructions and assessment requirements.
• Obtaining guidance on the organisation and structure of a scientific report.
• Formatting sections, headings, tables of contents, and appendices.
• Improving grammar, sentence structure, and overall presentation of the report.

#

How the Accuracy of AI Outputs Was Ensured

All statistical analyses, model development, diagnostic testing, forecasting procedures, interpretations, and conclusions were performed by the authors using R and RStudio. The outputs generated from AI were reviewed critically and compared with the course materials, lecture notes, and results produced directly from the statistical software. No statistical results, model outputs, forecasts, or conclusions were generated by AI. The authors take full responsibility for the analysis, interpretation, and final contents presented in this report.

7. REFERENCES

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